



VIDEO SURVEILLANCE STORAGE

Implementing A 2-Tier Storage Solution



Most Affordable

Long Term Retention

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Understanding how video surveillance infrastructures can benefit from implementing a 2-Tier video storage solution that provides for more affordable, scalable, video storage options.



FOREWORD

Welcome to this introduction of what I believe is a useful ebook on 2-tier video surveillance storage systems utilizing Linear Tape-Open (LTO) technology as the second tier. I'm Rich Gadomski, Head of Tape Evangelism at FUJIFILM, and I'm thrilled to be sharing with you some insights about this fascinating and increasingly relevant topic.

As you know, video surveillance has become an essential tool in many sectors of our society, from law enforcement to retail, transportation, and critical infrastructure. The ability to capture and analyze video footage has enabled organizations to enhance their security, prevent crimes, optimize operations, and make data-driven decisions.

However, with the proliferation of high-definition cameras, the exponential growth of video data, and the increasing demand for longer retention periods, managing video surveillance storage has become a significant challenge for many IT and security departments.

That's where the concept of 2-tier video surveillance storage systems comes in. Rather than storing all video data on expensive, energy intensive, high-performance storage devices such as hard disk drives, organizations can leverage a 2-tiered approach that provides for the quick access of the most recent video and the availability of all recorded video no matter when it was originally stored. This approach can significantly reduce storage costs, optimize system performance, reduce carbon footprint and simplify video data management.

But what is LTO technology, and why is it a useful component of a 2-tier video surveillance storage system? Linear Tape-Open, or LTO, is an open-format magnetic tape technology that was first introduced in 2000 by IBM, HPE, and Quantum. Since then, LTO has become a widely adopted standard for data backup and archiving, thanks to its high capacity, fast transfer rates, reliability, low energy consumption and low cost per gigabyte.

In a 2-tier video surveillance storage system, LTO serves as the second tier where older or less frequently accessed video data is replicated from the primary storage layer, such as hard disk arrays.



FOREWORD

By using LTO tapes, organizations can store massive amounts of video data in a cost-effective and scalable manner, while also ensuring data integrity and long-term retention.

LTO tapes can currently provide up to 18TB of capacity per cartridge and transfer rates of up to 400MB/s, making them ideal for large-scale video surveillance environments. And these capacities and speeds are already set to increase.

But LTO technology is not just useful for its cost, capacity, energy profile and speed. LTO also offers advanced features that are critical for video surveillance storage, such as data encryption, WORM (Write-Once-Read-Many) functionality, and LTFS (Linear Tape File System) support. With data encryption, organizations can protect their video data from unauthorized access or theft, while WORM functionality ensures that once data is written to the tape, it cannot be overwritten or erased, thus preserving the authenticity and legal admissibility of the data.

LTFS enables users to access, manage, and share video data on LTO tapes as if they were on a disk or a network drive, simplifying data access and reducing playback times.

In today's video surveillance infrastructure, where data volumes are growing exponentially, and regulatory and compliance requirements are becoming more stringent, 2-tier video surveillance storage systems utilizing LTO technology as the second tier are becoming very compelling.

This book will delve deeper into the technical aspects of 2-tier video surveillance storage systems utilizing LTO technology, covering topics such as system architecture, data replication, LTO versus hard disk only offerings, LTFS and more. It will also provide practical tips and best practices for designing, implementing, and managing such systems, based on real-world scenarios and use cases.

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Video Storage Challenges in a Video Surveillance Infrastructure

CHAPTER ONE



Why Do We Only Look At Single Tier Storage In Video Surveillance Infrastructures?

Video surveillance systems have become an essential tool in modern-day security, with an ever-increasing demand for high-quality video footage. The importance of high-quality video surveillance recordings cannot be overstated, as they serve as a valuable tool in crime prevention, investigation, and prosecution. Video surveillance has brought numerous benefits to society, making it an essential component of modern security systems.

Video Management Software (VMS) is an essential tool used in the video surveillance industry to record, store, and manage security surveillance footage. Traditional VMS technology has relied on hard disk-based storage systems to store that video footage. However, this approach has become increasingly costly as the amount of video data being generated continues to grow exponentially.

One of the primary reasons why hard disk-based storage systems are costly is their limited capacity. While hard disks have indeed improved in storage capacities over the years, they still have limitations. With the increasing amount of high-definition video being generated, storage capacity becomes a significant concern. The more cameras in the surveillance system, the larger the storage capacity required, and the higher the costs associated with it.

Another factor that adds to the cost of using hard disk-based only storage systems is their maintenance requirements. Hard disks are mechanical devices that are subject to wear and tear and have a limited lifespan. When hard disks fail, they must be replaced, and the video

footage stored on them is lost unless it has been backed up or the storage solution is a RAID (Redundant Array of Independent Disks) system, again a costly endeavour.

Additionally, hard disk-based storage systems have a higher risk of failure due to their mechanical components. A single hard disk failure can cause data loss or, worse, system downtime. This can result in significant associated costs, decreased video recordings, and additional costs for repairing or replacing the hardware.

Another cost associated with hard disk-based storage systems is their energy consumption. Hard disks require a significant amount of energy to operate as well as keep that equipment cool, and as the amount of storage required increases, so does the energy consumption. This results in higher electricity bills, which add to the overall cost of the system.

Larger hard disk-based storage systems require physical space to store the hardware. This can be a significant concern for organizations with limited space or those that want to minimize their physical footprint. Additionally, as the amount of storage required increases, so does the physical space required, which can become a significant expense.

As the amount of video data being generated continues to grow, organizations should consider alternative storage technologies that can offer greater storage capacity, better reliability, reduced energy consumption, and reduced maintenance requirements. By doing

so, organizations can reduce their overall costs and improve the efficiency and effectiveness of their video surveillance systems.

This is where the concept of a 2-tier video surveillance storage comes into play. In this book we will explore the overall video storage technologies that are currently implemented as well as discuss a viable robust 2-tier storage infrastructure.

Why Video Surveillance in the First Place

Studies have shown that the presence of surveillance cameras reduces the likelihood of criminal activity. Criminals are less likely to engage in unlawful activities in areas where surveillance cameras are present. They are aware that their actions are being recorded, which significantly increases the chances of being caught. As a result, video surveillance acts as a deterrent to potential offenders, helping to create a safer environment.

In the event of a crime, high-quality video surveillance recordings play a crucial role in the investigation and prosecution process. They serve as valuable evidence that can be used to identify perpetrators and gather critical information about the crime. Video footage can provide law enforcement officials with a detailed account of the crime scene, allowing them to retrace the steps of the perpetrator and identify any potential witnesses. High-quality video footage can also help to identify suspects by providing a clear image of their faces, clothing, and other identifying features.

Video surveillance can also improve the response time of law enforcement officials. With access

to real-time footage, they can quickly assess the situation and respond appropriately. This can help to minimize the risk of injury or loss of life in emergency situations, making it an essential tool for public safety.

In addition to its crime-fighting capabilities, video surveillance also brings numerous benefits to society. For instance, it can be used to monitor traffic flow, helping to reduce congestion and improve safety on our roads. Video cameras can also be used to monitor public spaces, ensuring that they are clean and well-maintained. This can help to improve the overall quality of life in our communities, making them more attractive places to live and work.

Moreover, high-quality video surveillance recordings can be used to improve workplace safety. Employers can use video cameras to monitor the behavior of their employees, ensuring that they follow safety protocols and use equipment correctly. This can help to prevent workplace accidents and reduce the risk of injury or death.

Another benefit of video surveillance is its ability to provide valuable insights into consumer behavior. Retailers can use video cameras to monitor their customers, tracking their movements and observing their shopping behavior. With advanced analytics made possible by use of Artificial Intelligence (AI), retailers can improve their store layout and product placement, increasing sales and improving customer satisfaction.

Finally, video surveillance can also be used to

protect the privacy of individuals. By monitoring public spaces, law enforcement officials can leverage AI toolsets and quickly respond to potential threats and ensure that everyone is safe.

With the increasing need for surveillance systems in various industries, the demand for storing video footage has also risen. However, the cost of implementing video surveillance storage can be expensive, and the retention times for video footage are often short. There are several reasons why video surveillance storage is costly to implement and why retention times are limited.

First, video surveillance storage requires large amounts of storage space. Modern surveillance systems can capture high-resolution video footage, which takes up significant amounts of storage space. For instance, a single high-resolution camera can capture up to 60GB of video data per day.

In addition, surveillance cameras are often deployed in large numbers, and storing footage from multiple cameras can also require vast amounts of storage capacity. This requirement for extensive storage space can result in expensive and energy intensive storage equipment, which increases the cost of implementing the video surveillance system.

Secondly, video surveillance storage requires high-speed data transfer rates. Surveillance footage needs to be transferred to storage devices in real-time to ensure that the system is continually capturing new footage. The transfer of large amounts of data can be challenging,

and systems require high-speed networks to support the transfer of data. The need for high-speed networks can result in expensive equipment and increased installation costs.

Thirdly, the cost of video surveillance storage is also affected by the type of storage medium used. Traditional storage mediums such as hard drives and solid-state drives can be expensive, consume a lot of energy and their limited lifespan can result in additional costs for replacement and maintenance.

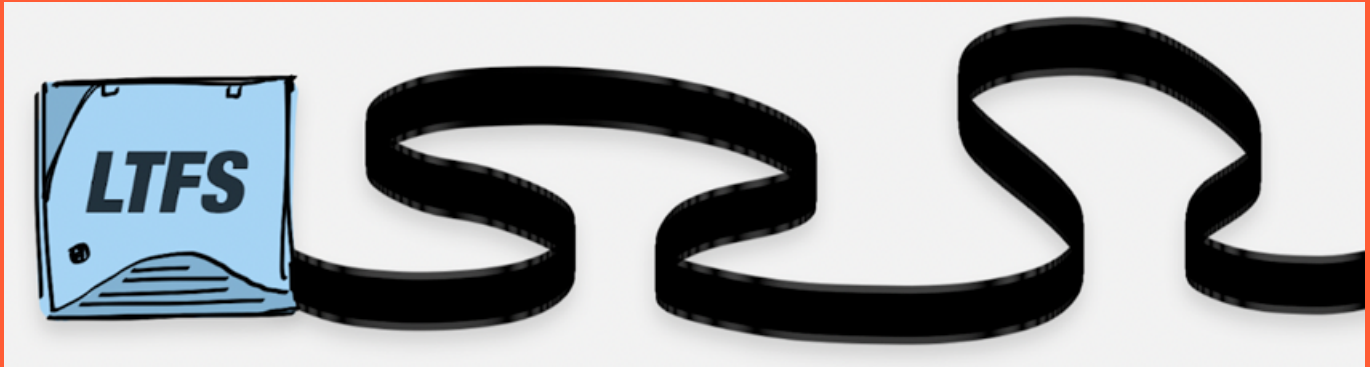
In addition, some industries require that video footage be stored for extended periods, which can require large-scale storage infrastructure, further increasing the cost of the storage solution. Advanced analytics based on AI also adds to storage requirements as the value of legacy video content increases.

In Chapter Three, we will discuss more ways in which the industry has attempted to deal with the increasing storage requirements by altering the video itself. We suggest that this approach is dealing with the wrong end of the problem. Concepts have been created including a reduction of the surveillance camera's resolution, frame rate, and bit-rate of the recorded video.

Instead, we need to look at the potential to utilize other proven storage technologies to mitigate the storage issue. The purpose of this book is to remove those mis-guided methods of dealing with the video storage conundrum and record all video at the best quality as to not compromise on safety and security.

CHAPTER TWO

What is Linear Tape-Open (LTO) And How Can It Be Used in Video Storage



As we know, video surveillance has become an integral part of modern society and as such there is a growing need for reliable and cost-effective storage solutions that can keep up with the demands of this ever-expanding industry.

The video surveillance industry as a collective needs to expand its view of proven and viable data (video) storage technologies and determine the most cost effective solution. One such solution is Linear Tape-Open (LTO) technology, which offers a wealth of benefits for video surveillance storage.

In this chapter, we will explore LTO technology in detail, including what it is, how it works, and its advantages for video surveillance storage. We will also take a look at the different generations

of LTO technology and their capabilities, as well as its compatibility with different video management software (VMS) applications.

Understanding LTO Technology

LTO (Linear Tape-Open) technology is a high-capacity, reliable, energy efficient and cost-effective tape storage solution that has become the industry standard for backup and archiving applications. Developed and supported by IBM, HPE, and Quantum, LTO technology has been around since the early 2000s and has constantly evolved through nine generations since its inception.

What is LTO Technology and How It Works

LTO technology uses a magnetic tape to store data, similar to the way audio cassettes and

VHS/VCR tapes worked in the past. However, unlike those formats, LTO tapes are specifically designed for digital data storage and ideal for modern video data recording, all with high-density recording, error correction, and other features that make them a perfect fit for video surveillance.

This is NOT Your Grandfather's Tape

Linear Tape-Open (LTO) and Video Home System (VHS) from VCR days are two different tape storage technologies that have been used in different as well as similar contexts, including video surveillance storage.

LTO is a magnetic digital computer data tape storage technology primarily designed for data protection and archiving, with a focus on high capacity, high speed, and data reliability. LTO tapes use a linear, serpentine recording pattern and have a data transfer rate of up to 400MB/s. LTO tapes are typically used for long-term data storage, and are known for their durability, reliability, low energy consumption, and low cost per terabyte.

On the other hand, VHS was a consumer-oriented analog tape recording format that was introduced in the 1970s for recording and playing back analog video signals. VHS tapes used a helical scan recording pattern, and had a maximum recording time of up to 6 hours (in the extended-play mode). VHS tapes had a lower resolution than today's modern digital storage media, and were subject to signal degradation over time due to now obsolete magnetic particle technologies that were employed in VHS tapes.

In terms of video surveillance storage, LTO tapes have been used for storing large amounts of video data for long periods of time, typically months or years. This is because LTO tapes offer high capacity, low cost and data reliability, and can be easily stored in off-site locations for disaster recovery purposes. In contrast, VHS tapes were analog and had limited capacity and lower quality compared to today's modern digital data tape storage media.

LTO tapes are housed in a rugged cartridge, which protects the tape from dust, dirt, and other contaminants that can damage it. When data is written to the tape, it is done so in a sequential fashion, with the tape moving past a read/write head that magnetically records and verifies the data integrity at the same time.

Advantages of Using LTO Technology for Video Surveillance Storage

The sequential nature of how video data is written to LTO tapes makes them similar to video surveillance recordings, which are also a sequential data stream. LTO tapes are written to and read from in a linear fashion, with data being written to the tape one block at a time in a continuous sequence. Similarly, video surveillance recordings capture data in a sequential stream of images, with each frame building on the one before it to create a complete video.

Similarly, the sequential nature of LTO tapes makes them an effective tool for recording, storing, and managing large amounts of video data. Because video data is written to the

tape in a continuous sequence (just as video is constructed), the video can be easily retrieved in the same order it was written. This is important for VMS systems to quickly retrieve and playback any recorded video.

The sequential nature of both LTO tapes and video surveillance recordings make them similar in how they store and retrieve data. This sequential data stream is what makes video surveillance recordings an effective tool for security and surveillance purposes, and LTO tapes an effective tool for data backup and archiving.

The Advancements of LTO via LTFS

Linear Tape File System (LTFS) is a file system that was introduced in 2010 to revolutionize the way data is stored on Linear Tape-Open (LTO) tape drives and media. LTFS was developed by IBM and is an open standard that enables users to write data to an LTO tape in a manner that is similar to writing to a hard drive or other storage device. LTFS has significantly improved the usefulness of LTO tape storage systems by making it easier to find, share and manage data on LTO tapes.

Before LTFS, LTO tapes were typically used for long-term storage, backup, and archiving purposes. However, accessing and managing data on these tapes could be a complex and time-consuming process. Users had to rely on specialized software and commands to retrieve data from tapes, which required specific training and expertise.

LTFS changed this by introducing a file system for LTO tapes that allows users to simply drag

and drop files onto the tape as if it were a removable hard drive or USB stick. This file system also enables users to browse the contents of the tape, search for specific files, and access files directly without having to restore the entire tape.

One of the major advantages of LTFS is its cross-platform compatibility. Since LTFS is an open standard, it can be used on a variety of operating systems and hardware platforms. This means that users can access and manage LTFS-formatted tapes on different devices and operating systems without the need for specialized software. It also simplifies the process of migrating data between tapes, which is particularly important for long-term storage and archiving purposes.

James Bain, a professor of electrical and computer engineering at Carnegie Mellon University, stated it very clearly when he said:

“The message about tape is that, just as SSDs did not displace HDDs, HDDs, in turn will not displace tape systems. All of them co-exist in a tiered storage architecture where cost and latency are balanced at every level of the system.”

The Advantages of LTO/LTFS in Video Surveillance Infrastructure

LTO technology offers several advantages for video surveillance storage, including high capacity, scalability, reliability, and cost-effectiveness and makes it an ideal choice as the second tier in a 2-tier video surveillance storage system:

Scalability: LTO technology has a roadmap that extends to at least 14 generations, providing predictable and scalable capacity growth. The latest LTO-9 technology can store up to 18TB of uncompressed data, which is sufficient for long-term storage of video surveillance footage. This allows for more video surveillance footage to be stored on a single high-density cartridge, reducing the need for multiple cartridges to store the same amount of video;

Reliability: LTO technology provides high reliability and data integrity. The tape drives and media are designed for long-term archival storage, with an expected lifespan of up to 50 years. Additionally, LTO technology includes built-in error correction and detection mechanisms, ensuring data integrity during storage and retrieval. Today's modern LTO tape has bit error rates that are several orders of magnitude better than HDD, SSD or optical devices;

Cost-Effective: LTO tapes are a cost-effective solution for storage, with a low cost per GB and low total cost of ownership compared to other storage options. This makes it an ideal choice for storing large amounts of video surveillance footage for extended periods of time;

Durability: Palm sized LTO tapes are designed to withstand harsh environments, including temperature changes, dust, and humidity. This ensures that the data stored on the tapes remains secure and accessible, even in challenging conditions;

High Transfer Rates: LTO technology offers high transfer rates, with the latest LTO-9 tape

offering transfer rates of up to 400MB/s. This allows for faster access to the video surveillance footage, making it easier to retrieve and playback recorded video;

Energy Efficient: Unlike HDDs that are constantly spinning 24/7 consuming a lot of energy and generating heat that needs to be cooled, tape cartridges only consume energy when being read or written by a tape drive. Otherwise, most of the time a tape cartridge is not consuming any power as it rests in a tape library slot or on a shelf in an offsite vault. For this reason, studies show that over time, tape systems consume 87% less energy than equivalent capacities of HDD. Over the product lifecycle including raw materials, production, distribution, usage and disposal, tape system produces an astonishing 97% less CO₂e than HDD. This helps organizations meet increasingly vital ESG goals in light of climate change and global warming;

Security: LTO technology offers advanced security features, including hardware-based encryption, to protect the data stored on the tapes. This is especially important for video surveillance footage, which may contain sensitive information that needs to be kept secure. WORM (Write Once Read Many) functionality is also available. Finally, LTO tape is compact and highly portable making it easy to enable an offsite, offline second copy with an air gap for prevention of unauthorized access via the network.

The LTO consortium, which includes IBM, Hewlett Packard Enterprise, and Quantum Corporation, has developed a roadmap for LTO

generations that defines the capabilities and features of each successive generation.

There are currently nine generations of LTO, with LTO-9 released in September of 2021. Here's a brief overview of each generation:

LTO-1: Introduced in 2000, LTO-1 had a storage capacity of 100GB and a transfer rate of 20MB/s.

LTO-2: Released in 2003, LTO-2 doubled the storage capacity of LTO-1 to 200GB and increased the transfer rate to 40MB/s.

LTO-3: Released in 2005, LTO-3 had a storage capacity of 400GB and a transfer rate of 80MB/s and WORM feature added.

LTO-4: Introduced in 2007, LTO-4 increased the

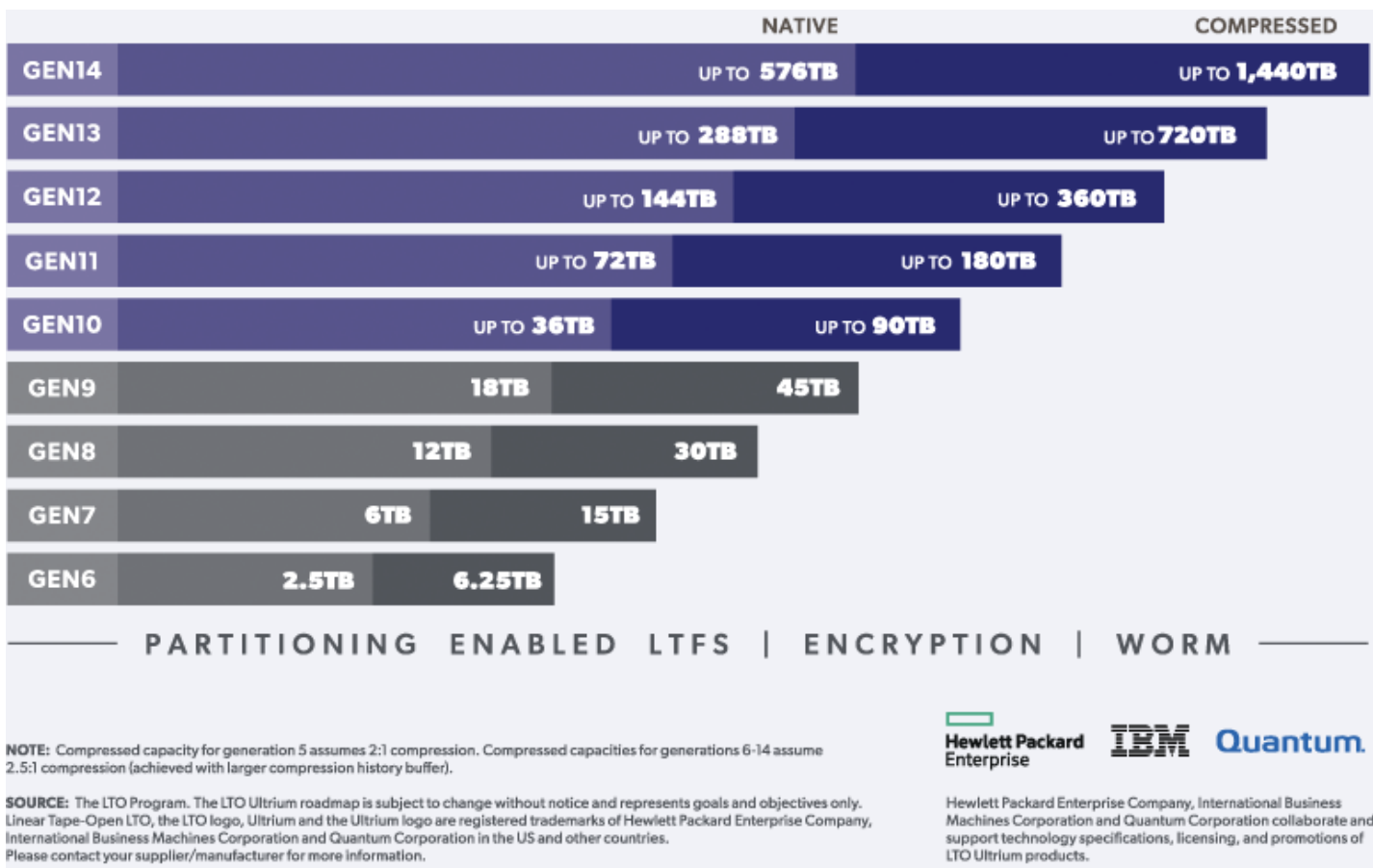
storage capacity to 800GB and the transfer rate to 120MB/s and drive level encryption added.

LTO-5: Released in 2010, LTO-5 had a storage capacity of 1.5TB and a transfer rate of 140MB/s and LTFS added.

LTO-6: Introduced in 2012, LTO-6 increased the storage capacity to 2.5TB and the transfer rate to 160MB/s and high performance Barium Ferrite metal particle introduced.

LTO-7: Released in 2015, LTO-7 had a storage capacity of 6TB and a transfer rate of 300 MB/s.

LTO-8: Introduced in 2017, LTO-8 doubled the storage capacity to 12TB and increased the transfer rate to 360MB/s.



LTO-9: Released in 2021, LTO-9 can store 18TB and a transfer rate of 400MB/s.

LTO-10: The tenth generation of LTO technology is currently under development, with expected native capacities of 36TB.

The LTO roadmap has already been updated to go out to a LTO-14 generation that is expected to house 576TB of data (read: video) on a single cartridge. Said another way, that is over ½ a Petabyte on one cartridge.

Each generation of LTO tape is backward compatible with the previous generation, meaning that LTO-8 drives can read and write to LTO-7 tapes, LTO-7 drives can read and write to LTO-6 tapes, and so on.

Therefore, investing in an LTO storage infrastructure allows for continuous upgrade capabilities and the ability to record, store, and manage larger and larger amounts of video data.

On the other hand, hard disk drives (HDDs) have also undergone significant advancements in their roadmap development. HDDs are the most widely used data storage solution and can be found in everything from personal computers to enterprise servers.

The first HDD was introduced in 1956 by IBM and had a capacity of 5 MB. Since then, HDDs have undergone significant advancements in terms of capacity, speed, and reliability.

The first major advancement in HDDs came in the 1980s with the introduction of the Winchester disk

Over the years, the storage capacities of HDDs have increased significantly, and they have become more reliable and efficient. However, as HDDs near their areal density limitations, it is not likely that they will deliver increasing capacities on the scale of LTO tape technology.

Here, we will discuss the roadmap of HDDs and the future development of various storage capacities.

Comparison: The roadmap of HDDs

The roadmap of HDDs is an essential aspect of the development of these storage devices. It outlines the future development plans for HDDs and sets the goals for the industry. The roadmap typically covers a period of five to ten years and is updated regularly to keep up with the latest technological advancements.

One of the key goals of the roadmap is to increase the storage density of HDDs. The storage density is the amount of data that can be stored on a single disk platter. Over the years, the storage density of HDDs has increased significantly.

In the 1980s, the storage density was around 1 megabyte per square inch. By 2020, the storage density had increased to around 1 terabyte per square inch. This increase in storage density has been made possible by the use of new technologies such as perpendicular magnetic recording (PMR) and shingled magnetic recording (SMR).

Another goal of the HDD roadmap is to increase the areal density of HDDs. The areal density is the amount of data that can be stored in a given area of the disk platter. To

increase the areal density, the industry has been working on reducing the size of the magnetic grains on the disk platter. This has been made possible by the use of heat-assisted magnetic recording (HAMR) and microwave-assisted magnetic recording (MAMR).

The future development of hard disk storage capacities

As technology advances, the storage capacities of HDDs are expected to continue to increase. The following are some of the future developments that are expected in the storage capacities of HDDs.

1. HAMR technology

HAMR technology is expected to be the next major advancement in the storage capacities of HDDs. It uses a laser to heat the magnetic grains on the disk platter, making them more susceptible to magnetic fields. This allows for smaller magnetic grains to be used, which increases the areal density of the disk platter.

2. MAMR technology

MAMR technology is another technology that is expected to increase the storage capacities of HDDs. It uses microwave energy to generate a magnetic field that is used to write data to the disk platter. This technology allows for smaller magnetic grains to be used, which increases the areal density of the disk platter.

3. SMR technology

SMR technology is another technology that is expected to increase the storage capacities of HDDs. It allows for overlapping tracks on the disk platter, which increases the areal density of

the disk platter. SMR technology is expected to increase the storage capacities of HDDs to around 20 terabytes per disk by 2025.

However, when SMR drives are used in a video surveillance system, they pose significant challenges when it comes to RAID rebuilds and reading data from such drives. RAID (Redundant Array of Independent Disks) is a technology that combines multiple HDDs into a single logical unit for data redundancy, performance improvement, or both.

In case of a disk failure, the data from the failed disk is reconstructed onto the remaining disks in the array, a process called RAID rebuild. However, RAID rebuilds can be challenging with SMR drives as their write pattern is different from conventional drives. SMR drives use overlapping tracks, meaning that the write tracks overlap each other, making it challenging to read and write data.

Furthermore, the decreased speed of reading data from SMR drives is another significant challenge in a video surveillance implementation. Reading from SMR drives can be slower due to the overlapping tracks, which result in longer access times. This can significantly affect the overall performance of the system and can cause bottlenecks.

Another challenge with SMR drives is their limited rewrite cycles. The overlapping tracks result in a higher number of rewrites, which can decrease the lifespan of the drive.

While SMR technology offers a higher data density, it poses significant challenges when it comes to RAID rebuilds and should be

avoided when using in a video surveillance recording systems.

4. SSD and HDD hybrid technology

Hybrid drives that combine HDDs and solid-state drives (SSDs) are another development that is expected to increase the storage capacities of HDDs. These hybrid drives use the large storage capacities of HDDs and the fast read and write speeds of SSDs. Hybrid drives are expected to increase

Solid state devices (SSDs) have become increasingly popular in the storage industry due to their fast access times, low latency, and high durability. However, in the video surveillance industry, using SSDs as the primary storage medium can present several challenges and drawbacks that can impact overall system performance and reliability.

One of the biggest challenges of using SSDs in a video surveillance storage infrastructure is cost. SSDs are significantly more expensive than traditional hard disk drives (HDDs) when it comes to cost per terabyte of storage. This can make it difficult to justify the expense of using SSDs, particularly in large-scale video surveillance deployments where terabytes or even petabytes of storage may be required.

Another challenge of using SSDs in video surveillance is their limited lifespan. Unlike HDDs, which can typically last for several years, SSDs have a finite number of write cycles before they begin to degrade. This is due to the fact that SSDs use flash memory to store data, which can only be written and erased a limited number of times. Over time, this can lead to data loss and system failure if not properly managed.

To address this challenge, many SSD manufacturers have developed technologies such as wear levelling and garbage collection, which help to prolong the life of SSDs by distributing writes evenly across the device and preventing unnecessary writes to areas of the drive that are already in use. However, these technologies are not foolproof and may not be enough to fully mitigate the risks associated with using SSDs in video surveillance storage.

Another drawback of using SSDs in video surveillance is their limited capacity. While SSDs have been increasing in capacity over the years, they are still not able to match the storage density of HDDs. This can make it difficult to achieve the same level of storage capacity using SSDs, particularly in larger-scale video surveillance deployments where large amounts of storage are required.

Furthermore, SSDs are more susceptible to temperature fluctuations than HDDs. SSDs generate more heat than HDDs, and this can lead to performance issues if the device becomes too hot.

Additionally, if the temperature drops too low, the SSD may become unstable or fail altogether. This means that video surveillance systems using SSDs may require additional cooling and heating infrastructure to ensure that the devices remain within their optimal operating temperature range.

Finally, SSDs have a lower tolerance for power fluctuations and surges than HDDs. While HDDs are designed to withstand power interruptions and fluctuations, SSDs can be

more sensitive to these events. This can lead to data corruption and system failure if the device is not properly protected against power-related issues.

To overcome the challenges associated with using SSDs in video surveillance storage, it is important to take a holistic approach to system design and management. This includes selecting SSDs with the appropriate capacity, reliability, and performance characteristics, as well as implementing advanced data management techniques such as data compression and deduplication to optimize storage efficiency.

While SSDs offer several advantages over traditional HDDs in terms of speed, durability, and performance, they also present several challenges and drawbacks when it comes to video surveillance

CHAPTER THREE

Describing the 2-Tier Video Surveillance Storage Concept



Video surveillance provide a reliable way to monitor and protect physical spaces. To meet the demands of storing such vast amounts of video, a 2-tier storage system utilizing Linear Tape-Open (LTO) technology has emerged as a viable and cost-effective solution.

Challenges with Traditional First-Tier Hard Disk Only Storage Solution

While HDD-based solutions have been the primary storage solution for video surveillance systems for many years, they come with several challenges. One of the main challenges is their limited capacity and high cost per terabyte of storage. Additionally, as data retention periods increase, the number of HDDs required to store this data also increases, resulting in increased power usage, heat generation, and cooling requirements.

Another significant challenge with HDD-based solutions is their susceptibility to data loss due to hardware failure or environmental factors. HDDs can be prone to failure due to mechanical issues, resulting in data loss or corruption. Additionally, environmental factors such as extreme temperatures, humidity, or magnetic fields can render HDDs unusable, resulting in data loss and costly recovery efforts.

The current storage solutions used in video surveillance systems are expensive and may not meet the requirements for large-scale systems. This has led to a need for more efficient and cost-effective storage solutions that can handle the massive amounts of data generated by video surveillance systems.

In dealing with these storage costs, technologists have come up with ways to reduce the amount of storage used by these surveillance cameras including altering the surveillance camera's resolution, frame rate, and bit-rate of the recorded video.

Resolution is the number of pixels that make up an image. A higher resolution camera captures more details in the footage, resulting in a higher-quality image. However, this also means that the video file size is more significant, and it requires more storage space. By reducing the resolution of the camera, it is possible to reduce the storage requirements without compromising the image quality too much.

Frame rate refers to the number of frames per second (fps) that the camera captures. A higher frame rate provides a smoother video, but it also requires more storage space. A lower frame rate reduces the storage requirements and can be a viable option if the footage does not require high-level detail or fluid motion.

Bit-rate refers to the amount of data the camera records per second. A higher bit-rate results in a higher-quality video, but it also requires more storage space. By reducing the bit-rate, the video's quality is reduced, but the storage requirements are significantly reduced.

In addition to the above, there are several other ways to reduce storage requirements while still maintaining the quality of the footage. One method is to use motion detection, which means that the camera only records when there is pre-determined amount movement in the frame. This reduces the amount of footage

captured and, in turn, reduces the storage requirements.

Record-on-motion video surveillance storage has become an interesting component of modern-day security systems. The technology is based on the concept of capturing video footage only when motion is detected. This method of surveillance has several advantages, such as reducing storage requirements and increasing the lifespan of the storage devices. However, it also has some significant drawbacks that are worth considering.

One of the noteworthy drawbacks of record-on-motion video surveillance storage is the possibility of missing crucial events. If the motion detection system is not properly configured, it may not capture some important events. For instance, if the system is set to detect motion only when the object is in a certain area, any activity outside the designated area will go unnoticed. Similarly, if the system's sensitivity is not appropriately calibrated, it may not detect some slow-moving objects, resulting in non-recorded video footage.

Another issue with record-on-motion video surveillance storage is the potential for false positives. When the system detects motion, it starts recording video footage. However, this can result in unnecessary footage of non-threatening events, such as trees swaying in the wind or animals moving around. These false positives can take up valuable storage space and make it difficult to find relevant footage during an investigation.

Record-on-motion video surveillance storage has its benefits, but it also has several drawbacks that need to be considered. Recognizing that really the only reason record-on-motion functionality exists in the first place is because of the storage requirements for continual video recording. If we mitigate the storage problem, why would we not record at full motion (30fps) or even higher, especially if the video surveillance camera that was purchased has the ability to record at 30fps or higher?

Reducing the surveillance camera's resolution (quality), frame rate, and bit-rate of the recorded video are all implemented in today's video surveillance storage systems to reduce storage requirements.

Interestingly, all of these methods either degrade or remove the captured video in one way or other and are all centered around the idea of saving storage space. This is at the added expense of the quality capabilities of the high-definition camera itself. Organizations purchase high-definition surveillance cameras in order to obtain the highest quality video available with today's technology, only to record that video in either missing or degraded quality.

The overarching purpose of this book is to remove those mis-guided methods of dealing with the video storage conundrum and record all video at the best quality as to not compromise on safety, security, and business value.

One of the main challenges of video surveillance storage systems is the sheer amount of data that they generate. For example, a single 4K camera can generate up to 12GB of data per

hour, which can quickly add up in a large-scale system. This requires a storage solution that can handle the massive amounts of data generated by these systems.

Another challenge is the need for scalability. As video surveillance systems grow in size, the storage requirements also increase. Traditional storage solutions such as hard drives may not be able to handle the growing storage needs of these systems. Therefore, a more scalable storage solution is needed to ensure that the system can grow without compromising on performance.

Introducing the Concept of 2-Tier Video Surveillance Storage

A 2-tier video surveillance storage system consists of two levels of storage media – a primary or online storage tier, which consists of high-performance HDDs or SSDs, and a secondary or nearline/offline storage tier, which comprises of LTO drives and high density LTO cartridges.

The primary storage tier is used to store the most recent and frequently accessed data, while the secondary tier is used to store all of the recorded video from all of the surveillance cameras.

This approach offers several advantages over traditional storage systems. First, it enables more efficient use of expensive primary storage by offloading infrequently accessed data to cheaper secondary storage.

Secondly, it allows for more extended retention periods, which is critical for investigations,

legal compliance, and/or safety reasons. Finally, it provides an additional layer of data protection and security by enabling offline storage of a second copy of critical surveillance footage.

Overview of Video Management Software Applications

Video management software (VMS) applications are central to modern video surveillance systems, enabling the capture, storage, and retrieval of video footage. These applications provide a user interface to manage cameras, monitor live feeds, and search and playback recorded footage, primarily through a graphical user interface concept of a timeline.

A critical function of video management software applications is to manage the data lifecycle of recorded video footage. This includes retention policies, data movement between storage tiers, and ultimately video deletion. The software must ensure that data is stored in the appropriate tier, considering factors such as access frequency, retention requirements, and storage costs.

How the VMS Must Be Aware of LTO Technology to Properly Use 2-Tier Storage

Video management software (VMS) applications must be aware of the storage technology used in the 2-tier storage system to ensure efficient and effective use of the storage resources. In the case of LTO technology, the software must understand how the LTO drives operate, the media capacity, and the write and read speeds of the LTO drives. This knowledge enables the software to determine when to move data from the primary to the secondary storage tier and how to quickly identify and playback any recorded video from the VMS timeline.

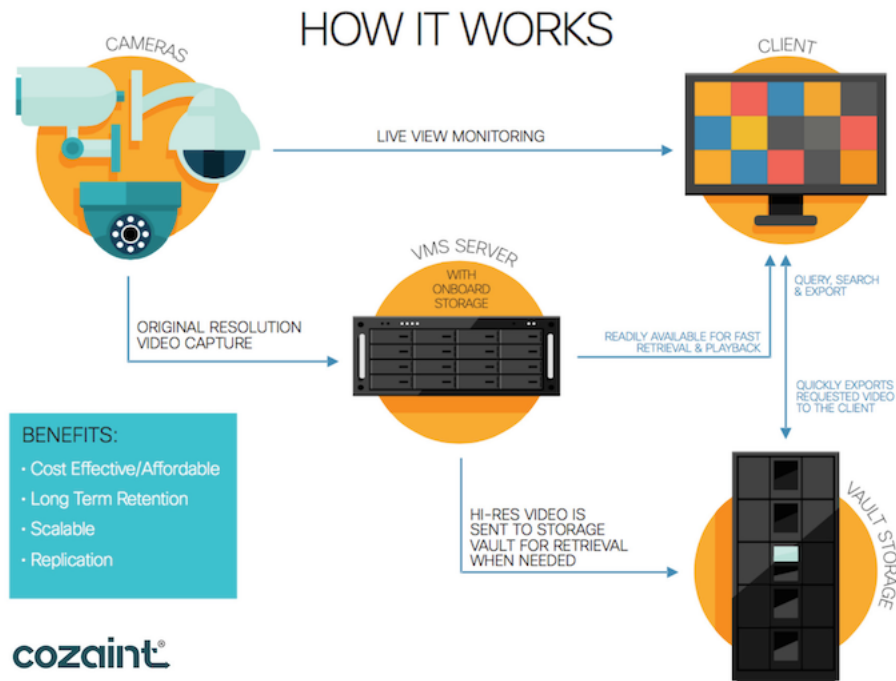
The VMS software must also be capable of managing the data movement between the primary and secondary tiers seamlessly. This includes transparently replicating video from the primary tier to the secondary tier when it becomes less frequently accessed

A real-world example of the overall costs associated with traditional hard-disk based video surveillance storage systems could be a large retail chain with hundreds of stores that require video surveillance for security purposes.

In such a scenario, the company may need to install hundreds of cameras at each store, which would require a significant amount of storage space to store the video footage. Depending on the resolution and frame rate of the cameras, a single day's worth of footage for a single camera could take up several gigabytes or even terabytes of storage space.

To store all the video footage, the company would need to purchase a significant amount of hard-disk based storage systems, which can be expensive. In addition to the initial cost of purchasing the storage systems, there would also be ongoing costs associated with maintaining and upgrading the systems as needed.

The company would also need to consider the cost of electricity to power the storage systems, as well as the cost of cooling the systems to prevent overheating. Over time, the cost of electricity and cooling can add up and become a significant expense. Related CO2 emissions from this power consumption are concerning as well.



Furthermore, the company may also need to consider the cost of personnel to manage and monitor the storage systems, as well as the cost of any necessary software licenses or maintenance contracts.

Calculating Total Cost of Ownership

Total Cost of Ownership (TCO) becomes a financial sustainability issue, in many cases reducing the retention times again for such systems. Overall, the costs associated with traditional hard-disk based video surveillance storage systems can be substantial, particularly for larger organizations with multiple locations and a high volume of video footage to store.

The need for a more cost-effective and scalable storage solution in video surveillance storage systems is crucial to ensure that these systems

can handle the massive amounts of data generated by them.

The Need for 2-Tier Storage in a Video Surveillance Storage Infrastructure

In video surveillance systems, data retention is critical to ensure that critical evidence is not lost. The longer the retention period, the more data is generated, and the harder it is to manage and store. Typically, video surveillance storage infrastructure is divided into two tiers: the first tier is used for short-term storage, and the second tier is used for long-term storage.

The first tier is typically composed of hard disk drives (HDDs) or solid-state drives (SSDs) and is used for storing video footage that has been recently captured. The second tier is used for storing video footage for longer retention

periods and can be composed of tape based storage solutions, such as LTO.

LTO-based storage solutions are much more cost-effective than HDD-based solutions, offering a significantly lower cost per terabyte of storage. Additionally, tape-based solutions can store data for much longer periods of time without the need for additional hardware or maintenance, consumes far less energy, further reducing the total cost of ownership.

LTO technology is highly reliable, with an error rate of only one in every 10^{20} bits. Additionally, LTO tapes can withstand environmental factors on par with HDDs, such as a wide range of temperatures, humidity, and magnetic fields. This reliability ensures that data is not lost due to hardware failure or environmental factors, reducing the risk of data loss and the cost associated with recovery.

Known Savings to Electricity Usage, Cooling Systems, and Maintenance Using LTO

In addition to the cost savings associated with LTO technology, utilizing tape-based storage solutions also offers significant savings in electricity usage, cooling systems, and maintenance costs. Tape-based storage solutions do not require any power when not in use, unlike HDDs or SSDs, which require constant power to maintain their data. This results in lower electricity usage, reducing energy costs and carbon footprint.

In today's environmentally friendly technology and sustainability push, an LTO-based 2-tier video storage solution is a leading "Green" solution.

CHAPTER FOUR

Best Practices for Implementing a 2-Tier Video Surveillance Storage Solution



Tiered Storage Solutions

In this chapter, we will discuss the best practices for implementing and maintaining a 2-tier video surveillance storage from the perspective of the video management system (VMS) software and explain why it is critical for the VMS to understand an LTO 2-tier storage infrastructure.

We have seen in recent years the various uses of video surveillance systems and the tremendous growth of these systems, which has resulted in an explosion of video data that needs to be recorded, stored and managed.

As we have highlighted already, video storage is one of the critical components of a video surveillance system and it needs to be designed in a way that ensures data availability, reliability, scalability, and easy video playback. One solution

that is gaining viability in recent years is the 2-tier storage architecture utilizing Linear Tape-Open (LTO) technology.

When designing a 2-tier storage architecture utilizing LTO, several factors need to be considered. These include the required storage capacity, the expected data growth rate, and the retention period of the data. The secondary storage tier's capacity should be sufficient to store video for the desired retention period and it should be easily expandable to cater to future video retention needs.

The volume of video footage generated by these cameras is enormous, making it critical to have an efficient storage system that can handle the data load. A 2-tier storage infrastructure, comprising high-performance

disk-based storage for short-term storage and Linear Tape-Open (LTO) tape-based storage for long-term retention, which has emerged as an effective solution for video surveillance storage.

Understanding the 2-Tier Storage Infrastructure

A 2-tier storage infrastructure consists of two types of storage media: high-performance disk-based storage and LTO tape-based storage. The high-performance disk-based storage is used for short-term storage, typically for a few days to a few weeks, while the LTO tape-based storage is used for long-term retention, usually for several months or even years.

The high-performance disk-based storage is expensive, but it offers fast access to data, making it ideal for storing video footage that requires quick retrieval, such as recent incidents or events. The LTO tape-based storage, on the other hand, is cheaper, more energy efficient, and offers more affordable higher-capacity storage, making it ideal for storing large amounts of video footage for a long time.

Best Practices for Implementing and Maintaining a 2-Tier Storage Infrastructure:

Choose the Right VMS Software

Choosing the right VMS software is critical when implementing a 2-tier storage infrastructure. The VMS software must be compatible with the storage system and provide seamless integration between the high-performance disk-based storage and the LTO tape-based storage. The VMS should also be able to manage the data lifecycle effectively, including replicating,

retrieving, and restoring data, to ensure data integrity and easy video playback.

Understanding the VMS Timeline and Why It Is The Key Component

A Video Management System (VMS) is an application that allows users to manage and record video from security cameras or other surveillance devices. One of the key features of a VMS application is the video playback timeline functionality, which allows users to review recorded video footage.

The video playback timeline is a graphical representation of the recorded video footage, displaying the time axis on the horizontal axis and the video frames on the vertical axis. The timeline is divided into segments of equal length, usually ranging from a few seconds to several minutes, depending on the settings of the VMS application. Each segment of the timeline represents a block of recorded video footage, and the user can navigate through the timeline to select the segment of interest.

When the user selects a segment of the timeline, the VMS application loads the corresponding video footage and displays it in the video playback window. The user can then play, pause, fast-forward, or rewind the video to view the footage in detail. The video playback window may also display additional information, such as the camera name, date and time of the recording, and any other metadata associated with the video footage.

The video playback timeline functionality is particularly useful for reviewing recorded video footage in cases of security incidents

or other events. For example, if a theft occurred in a store, the store owner can use the video playback timeline to navigate to the time when the incident occurred and review the footage to identify the perpetrator. Similarly, if a security breach occurred in a facility, the security personnel can use the video playback timeline to review the footage and determine how the breach occurred and who was responsible.

In addition to reviewing recorded video footage, the video playback timeline functionality can also be used to export video footage for further analysis or evidence purposes. The user can select the segment of the timeline that contains the relevant footage and export it to a video file or other format that can be shared with law enforcement or other authorities.

Overall, the video playback timeline functionality is a critical component of a VMS application, enabling users to efficiently review and analyze recorded video footage. The ability to quickly navigate through the timeline, select specific segments of footage, and export footage for further analysis makes the VMS application an essential tool for security and surveillance applications.

An ideal process for video playback from a VMS timeline perspective would be for the video operator -whom is needing to review video- to move the VMS timeline to the suspected date/ time and playback that video.

In a 2-tier storage infrastructure, the most recent video (for example, the last two weeks of recorded video) would be located on high-performance spinning hard disk storage. If the

video needed by the video operator was located on the first tier of storage, the video would be recalled and video playback would commence.

If the video that was requested by the video operator was further back in time (i.e., longer than two weeks ago) the 2-tier storage system would know on which LTO cartridge that video was stored on the and the LTO library would pull the cartridge from its holding slot within the library and insert the cartridge into the LTO drive where the system would seek to the appropriate time of video and copy it back for the operator to review.

This LTO video retrieval process would be approximately three to eight minutes of elapsed time before the video operator was actually viewing recorded video. And this is all without any other steps or (I.T.) personnel involved.

Properly sizing the first and second tiers of video data storage -and setting the expectations with the video operator- will deliver an ideal infrastructure to playback and review any and all recorded video either instantaneously or within an eight minute process.

Most video operators are amenable to those expectations and the business managers are able to realize a 40% to 60% cost savings over traditional video surveillance storage systems.

Use RAID for High-Performance Disk-Based Storage

Using redundant array of independent disks (RAID) for high-performance disk-based

storage can improve data availability and reduce the risk of data loss. RAID can also improve disk performance and increase the storage capacity by combining multiple disks into a single logical unit.

Use LTO-Based Storage for Long-Term Retention

Using LTO-based storage for long-term retention is an effective way to reduce storage costs and improve data reliability. LTO cartridges are designed for long-term storage and can withstand a variety of environmental conditions, making them ideal for storing data for an extended period.

Why VMS Software Needs to Understand LTO 2-Tier Storage Infrastructure

A VMS software that understands LTO 2-tier storage infrastructure can effectively manage video footage and ensure data integrity and security. Here are some reasons why VMS software needs to understand LTO 2-tier storage infrastructure:

Efficient Data Management

A VMS software that understands LTO 2-tier storage infrastructure can effectively manage the data lifecycle, including replicating, retrieving, and playing back video data. The software can automatically replicate data from high-performance hard disk based first tier to the more affordable LTO-based second tier of storage.

Integration of Video Management Software (VMS) Applications with LTO Technology

The integration of Video Management Software

(VMS) applications with LTO technology is critical for efficient and reliable storage of video data. A VMS application is used to manage and control the video surveillance system, including recording, playback, and analysis of video data. VMS applications need to be integrated with the 2-tier storage architecture to enable efficient and reliable data storage and retrieval.

The integration of VMS applications with LTO technology can be achieved in various ways. One way is through the use of a middleware layer that provides a standardized interface between the VMS application and the storage system. The middleware layer translates VMS commands into storage system-specific commands, enabling seamless integration and communication between the two systems.

Another way of integrating VMS applications with LTO technology is through the use of an Application Programming Interface (API). The API provides a set of standardized functions that enable the VMS application to communicate with the storage system directly. The API provides a higher level of integration and customization and can be used to develop customized applications that cater to specific storage needs.

While LTO storage technology provides an ideal solution for long-term video retention in VMS applications and despite the overwhelming benefits of Linear Tape-Open (LTO) storage for long-term video retention, many VMS solutions lack support for LTO storage.

A potential reason for the lack of LTO support in VMS solutions is the universal adoption of hard disk storage system such as Network Attached Storage (NAS) and Storage Area Networks (SAN) systems. VMS software vendors have not had the incentives to support LTO as the actual end-customers lack the awareness and appreciation of the benefits that LTO brings to a video surveillance storage infrastructure to therefore ask for *-or demand-* such a 2-tier storage solution.

Despite the challenges of implementing LTO storage, there are several solutions that VMS providers can explore to overcome these obstacles. One solution is to offer a 2-tier storage model that combines LTO storage with disk-based storage. This approach enables organizations to take advantage of the long-term retention benefits of LTO storage while providing faster access to data through disk-based storage solutions.

As the demand for video surveillance continues to grow, VMS providers must continue to explore new storage solutions that can meet the needs of their customers while providing a cost-effective, eco-friendly, and scalable solution.

An example of a professional-grade enterprise level VMS that fully supports LTO in a 2-tier video surveillance solution is Cozaint's askALICE. <https://cozaint.com/askalice/> askALICE video surveillance system can provide scalable long-term video retention at significantly lower costs over traditional hard-disk only storage retention systems.

CHAPTER FIVE

A Deeper Dive into LTO Technology



In today's world, data is a valuable resource, and the ability to store and access large amounts of it quickly and efficiently is critical. Scalable Linear Tape-Open (LTO) libraries provide a reliable and cost-effective solution for businesses that need to manage large amounts of data. In this chapter, we'll explore how scalable LTO libraries work, the amount of data storage capacities that can be obtained, and the cost benefits of these libraries.

LTO technology was first introduced in 2000 as an open-format tape storage solution for backup and archiving. The format was designed to offer a high level of data reliability and durability while providing high-capacity storage at an affordable price. The LTO Consortium, a group of technology companies, manages and develops

the LTO standard, which is currently in its ninth generation.

LTO libraries are made up of multiple tape drives, which are used to read and write data to LTO tapes. These tapes are contained in cartridges and they are kept in slots or bays within the library. As data is written to the tapes, the library software keeps track of the location of each piece of data, making it easy to retrieve when needed. LTO libraries can be scaled up by adding additional tape drives, cartridges, and storage slots and bays to meet growing data storage needs.

One of the significant advantages of LTO libraries is their scalability. LTO libraries can be scaled up to store and manage multiple

petabytes of video data, making them ideal for large enterprises and organizations. This scalability also allows businesses to start with a smaller library and add more tape drives, slots, and cartridges as their storage needs increase. LTO libraries can also be configured to automatically move data between tapes and disk storage, optimizing the use of storage resources.

The amount of data storage capacities that can be obtained with LTO libraries is impressive. The current generation of LTO tapes, LTO-9, can store up to 18TB of uncompressed data on a single cartridge. With a library containing 100 tape cartridges, an organization can store up to 1.8PB of uncompressed data. As LTO technology continues to evolve, storage capacities are expected to increase, providing even more storage options for businesses.

Cost Benefits of a 2-Tier Storage Architecture Utilizing LTO Technology

The cost benefits of a 2-tier storage architecture utilizing LTO technology are significant. The primary storage tier, which is typically based on high-speed disks, is expensive to implement and maintain. The secondary storage tier, which is based on LTO tape technology, is significantly more cost-effective, enabling organizations to store large volumes of data at a lower cost.

The cost of LTO tapes is also significantly lower than that of high-speed disks, making it an ideal storage medium for long-term video retention purposes.

The storage capacity of a LTO tier can be easily expanded by adding additional tapes to the

system, making it easy to cater to future data growth requirements. This scalability is especially beneficial to organizations that have rapidly growing data storage needs, as it enables them to add storage capacity without incurring significant costs.

Another cost benefit of LTO libraries is their low power consumption. LTO storage is an energy-efficient solution compared to other storage options, requiring less power to operate and cool. As data storage needs increase, the cost of powering and cooling storage systems can become a significant expense for businesses. LTO libraries can help reduce these costs by providing a more energy-efficient storage solution.

LTO libraries are compatible with a wide range of operating systems and software, making them a versatile storage solution. This flexibility allows businesses to select software that best meets their specific data management needs, rather than being locked into a proprietary software solution.

The Largest Linear Tape-Open (LTO) Library

The largest LTO libraries available from manufacturers such as IBM, HPE, Quantum, Spectra Logic, and Qualstar can contain hundreds or even thousands of tape drives and slots, depending on the specific model and configuration.

Here are some examples of the largest LTO libraries available from these manufacturers:

IBM TS4500: This is one of the largest LTO libraries available, capable of supporting up to 128 tape drives per library and up to 3,592 tape cartridges. The library can be configured with either LTO-8 or LTO-9 tape drives, and can store up to 1.04Exabytes 2.5:1 compressed data (video) per library.

HPE StoreEver MSL6480: This LTO library can support up to 42 tape drives and up to 560 tape cartridges. It can be configured with LTO-9 tape drives, and can provide up to 25.2Petabytes of uncompressed storage capacity.

Quantum Scalar i6000: This LTO library can support up to 192 tape drives and up to 12,006 tape cartridges. It can provide up to 1.8Exabytes of compressed storage capacity.

Spectra Logic T-Finity: This LTO library can support up to 144 tape drives and up to 56,400 tape cartridges. It can provide up to 1 exabyte of uncompressed data in a single tape library.

Qualstar XLS-832700: This LTO library can support up to 32 tape drives and up to 700 tape cartridges. It can be configured with LTO-9 tape drives, and can provide up to 4.38PB storage capacity.

It's important to note that these are just examples of the largest LTO libraries available from these manufacturers, and that other configurations and models may be available with different capacities and capabilities and manufacturers. Additionally, the number of drives and slots that can be supported may depend on the specific configuration of the library.

The Smallest LTO Library

The smallest LTO library typically consists of a single LTO drive and a few cartridge slots, usually between eight to 24 slots. Such a small LTO library is often referred to as “rack mountable” or as an “autoloader,” which means it can automatically load and unload tapes without requiring manual intervention.

The most appropriate small LTO library to use in a video surveillance storage infrastructure would depend on various factors such as the number of surveillance cameras in use, amount of video data being generated (video resolution), the retention period, the average amount of motion happening over a 24-hour period (H.264 recording specification), and the number of concurrent streams.

Therefore, it is important to realize the scalability of a 2-tier video storage solution when integrating LTO as the second tier. One could scale the LTO library from a couple of LTO drives to 144 LTO drives and from a handful of slots within a 2U library all the way up to tens of thousands of cartridges, delivering the most economical video storage solution on the market.

From these variables, one could calculate the amount of storage needed and design an appropriate 2-tier video surveillance storage solution.

From those variables, one could calculate the amount of storage needed and design an appropriate 2-tier video surveillance storage solution.

You can visit the FUJIFILM video storage calculator to get an ideal representation for the amount of video surveillance storage needed:

<https://www.fujifilm.com/us/en/business/data-storage/resources/video-surveillance-retention-cost-calculator>

Although a single LTO drive can provide sufficient performance for a small number of concurrent video streams, best practices for a 2-tier video storage architecture would utilize two LTO drives for those times when video playback is requested by the VMS operator and the continuation of recording of all surveillance cameras.

However, for larger video surveillance installations, a larger LTO library with multiple drives and hundreds of slots may be necessary to accommodate the vast amounts of video data generated.

In each scenario, the implementation of a 2-tier storage solution will cost significantly less expensive than a 1-tier hard disk only storage solution and save 40-60% of the storage budget.

Monitoring and Maintaining 2-Tier Storage Performance

Monitoring and maintaining the performance of 2-tier storage is critical to ensuring the integrity and availability of video surveillance data. The following are some key techniques for monitoring and maintaining 2-tier storage performance:

Regular Maintenance: Periodic maintenance of LTO tape drives, such as cleaning and calibration,

can ensure that they are functioning at their optimum performance levels.

System Monitoring: By monitoring the performance of the entire storage system, administrators can proactively identify and address any issues that may impact the availability or integrity of data.

With the increasing demand for longer retention times and larger-scale data storage requirements, a 2-tier storage architecture utilizing LTO technology is becoming a more viable solution.

By utilizing data management and optimization techniques such as data tiering, administrators can maximize the use of available storage capacity. Monitoring and maintaining the performance of 2-tier storage is critical to ensuring the integrity and availability of video surveillance data. With regular maintenance, system monitoring, and performance optimization, administrators can ensure that their storage infrastructure can handle tremendous amounts of video data that is generated by a multitude of video surveillance cameras.

CHAPTER SIX

Real World Examples of LTO Storage in Video Surveillance



In this chapter, we will explore real-world examples of organizations successfully implementing 2-tier storage and the benefits and ROI achieved through this technology.

Real-World Examples

The 2-tier video surveillance storage architecture is a viable and efficient solution for storing and managing video data. Here are a couple of real-world examples of implementing 2-tier video surveillance storage using LTO technology:

1. New York City Police Department:

The NYPD has one of the largest video surveillance systems in the world, with over 18,000 cameras deployed across the city. The NYPD implemented a 2-tier video surveillance storage solution using LTO technology to manage its massive video data. The active tier consists of

high-performance storage, including SSDs and HDDs, that are used to store video data for a short period of time. The cold tier consists of LTO tapes that are used to store video data for long-term retention. The NYPD estimated that the 2-tier storage architecture using LTO technology reduced the cost of storage by over 70% compared to traditional storage solutions.

2. NASA:

NASA's Kennedy Space Center implemented a 2-tier video surveillance storage solution using LTO technology to manage the massive amounts of video data generated by its facilities. The active tier consists of high-performance storage, including SSDs and HDDs, that are used to store video data for a short period of time. The cold tier consists of LTO tapes that are used to store video data for long-term

retention. The LTO tapes are stored in an off-site location for disaster recovery purposes. The two-tier storage architecture using LTO technology enabled NASA to store and manage its video data efficiently and cost-effectively.

3. University of Oregon:

The University of Oregon implemented a two-tier video surveillance storage solution using LTO technology to manage the video data generated by its campus-wide surveillance system. The active tier consists of high-performance storage, including SSDs and HDDs, that are used to store video data for a short period of time. The cold tier consists of LTO tapes that are used to store video data for long-term retention. The university estimated that the 2-tier storage architecture using LTO technology reduced the cost of storage by over 50% compared to traditional storage solutions.

4. City of Oceanside CA Waste Water Plant

For the City of Oceanside the importance of long-term video retention was not only regulatory but also necessary for the security of the facility. Keeping a city's water treatment facility safe and secure is a serious matter to numerous stakeholders. Therefore, these surveillance cameras are 1080p high-def IP-cameras and are recording 24/7. Needing to work within the set City budget allotments, the City's I.T. team needed a solution that would deliver the best video recording and at the same time meet the appropriate state's record retentions requirements.

See Appendix 1 for the complete white paper on the City of Oceanside's Waste Water Plant's implementation of a **Cozaint askALICE** 2-tier video surveillance solution.

2-Tier Video Storage ROI

The ROI achieved through 2-tier storage using LTO technology is significant. Organizations that implement this technology can expect to see a return on their investment within a few years. The cost savings associated with LTO technology can be significant, especially for organizations that generate large amounts of video surveillance data. In addition, the scalability and reliability of LTO technology provide organizations with a long-term storage solution that will continue to provide value for years to come.

2-tier video surveillance storage using Linear Tape-Open (LTO) technology is a cost-effective and scalable solution for organizations that generate large amounts of video surveillance data. Real-world examples have shown that organizations can achieve significant cost savings while also improving their storage capacity and data integrity.

CHAPTER SEVEN

What About The Cloud?



Cloud-based storage has emerged as an attractive option for video surveillance infrastructures due to its flexibility, scalability, and initial cost-effectiveness. However, while the benefits of cloud-based storage for video surveillance are often touted, the challenges and limitations that come with it are not widely discussed. In this chapter, we will examine the challenges associated with using cloud-based storage in a video surveillance infrastructure, including the necessary network bandwidth, network costs, pricing models, ownership issues, and the costs of retaining more and more video each day.

Cloud companies offer various pricing models for video data storage, which can be confusing and complex. Some companies charge per gigabyte or per terabyte of data stored, while others

charge based on the number of cameras or users accessing the data. Additionally, cloud companies may also charge egress (download) fees for downloading or accessing data, which can add significant costs for customers. Egress fees can range from a few cents per gigabyte to several dollars per gigabyte, depending on the cloud provider and the location of the customer.

Another challenge associated with using cloud-based storage in a video surveillance infrastructure is the ownership issues of the data once stored in the cloud. Customers must be aware of who owns the data, how it can be accessed, and what rights they have to the data.

Some cloud providers may claim ownership of the data once it is uploaded to their servers, while others may provide customers with more control over their data. Customers must carefully review the terms and conditions of the cloud provider's service agreement to understand who owns the data and what rights they have to it.

Customer Network Bandwidth Connection

One of the most significant challenges of using cloud-based storage in video surveillance infrastructure is the necessary network bandwidth connection required to send video data to the cloud. The required network bandwidth connection is determined by many factors including the resolution and frame rate of the video. For example, a 1080p video recorded at 30 frames per second (fps) requires a network bandwidth connection of approximately 20Mbps, while a 4K video recorded at 60 fps requires a network bandwidth connection of approximately 100Mbps.

Video data is typically bandwidth-intensive, and uploading large volumes of data to the cloud can quickly consume local company network bandwidth capacity and slow down other critical network operations. Moreover, cloud-based storage is heavily dependent on internet connectivity, which means that any disruption in connectivity can significantly impact the reliability of the surveillance system.

To support a high-throughput high-bandwidth customer connection, businesses typically turn to fiber-optic connections. Fiber-optic connections offer high-speed, low-latency connectivity and are capable of delivering speeds of up to 1Gbps or more. However, the cost of such connections

can vary depending on the location of the business, the level of service required, and the provider.

Customer Costs

Implementing a cloud-based storage infrastructure for a video surveillance system can involve various costs for the customer. Here are some of the most significant costs to consider:

Network Connection Costs: The cost of the customer's network connection is the first and most important cost to consider when implementing a cloud-based storage infrastructure for a video surveillance system. The cost of the network connection will depend on the internet service provider and the type of connection required (fiber, cable, DSL, etc.).

Specifically, cost of a dedicated 1Gbps fiber internet connection can range from \$500 to \$1800 per month, with installation costs ranging from \$99 to \$500. It's essential to consider the total cost of ownership when evaluating options for a cloud-based storage system in a video surveillance infrastructure.

And if the customer has a multitude of high-definition surveillance cameras, a 1Gbps network connection may not be big enough to handle this upload traffic.

According to a survey by the Fiber Broadband Association, the average cost of a dedicated 1Gbps fiber internet connection for a business in the United States is around \$1,000 to \$2,000 per month, with installation costs ranging

from \$5,000 to \$10,000. However, these costs can vary significantly depending on the location and the provider.

According to a report by BroadbandNow, the average cost of a 1Gbps fiber-optic connection for businesses in the United States is \$1,057 per month. However, the cost can vary significantly depending on the location, with some cities offering significantly lower prices. For example, in Kansas City, Missouri, businesses can purchase a 1Gbps connection for as little as \$70 per month.

Cloud storage providers offer various pricing models for video data storage, including pay-as-you-go, tiered pricing, and flat-rate pricing. Pay-as-you-go pricing is based on the amount of video data stored and the duration of storage. Tiered pricing is based on the volume of video data stored, with higher volumes attracting lower rates. Flat-rate pricing is based on a fixed monthly or annual fee, regardless of the volume of video data stored. Customers should carefully evaluate pricing models to ensure they select the most cost-effective option for their video surveillance infrastructure.

Most providers offer several tiers of service, with higher tiers providing faster speeds and more reliable connections. For example, Comcast Business offers a range of service tiers, with prices ranging from \$69.95 per month for a 100Mbps connection to \$499.95 per month for a 10Gbps connection.

Here are some examples of costs for high-throughput, high-bandwidth customer connections from typical business internet providers (at the time of this writing):

Comcast Business: Comcast Business offers a range of high-speed internet plans for businesses, including a 1Gbps plan with symmetrical upload and download speeds. According to their website, the cost for this plan starts at \$499.95 per month, with a three-year agreement and a \$500 installation fee.

AT&T Business: AT&T Business offers several high-speed internet plans, including a 1Gbps plan with symmetrical upload and download speeds. According to their website, the cost for this plan starts at \$1500 per month, with a two-year agreement and a \$99 installation fee.

Verizon Business: Verizon Business offers a range of high-speed internet plans for businesses, including a 1Gbps plan with symmetrical upload and download speeds. According to their website, the cost for this plan starts at \$1799.99 per month, with a two-year agreement and a \$399 installation fee.

It's worth noting that these costs may not include other expenses such as equipment, maintenance, and support, which can add to the overall cost of a high-throughput, high-bandwidth customer connection.

Retaining more and more video each day becomes a costly proposition. As video surveillance systems become more advanced and cameras capture higher-quality video, the amount of data generated can quickly grow.

Let's say a customer is using 50 cameras to monitor their business, and each camera captures footage at a resolution of 1080p. Assuming that the footage is being stored for

30 days, the customer will need to upload a total of 16.2 Terabytes of data to the cloud each month. According to Amazon Web Services (AWS), the cost of transferring 16.2 Terabytes of data to the cloud each month would be approximately \$2,430. This cost does not include any other fees that the customer may incur, such as data retrieval or storage costs.

Cloud Storage Costs

After those initial customer network bandwidth connection charges, the customer will then need to pay the cloud storage provider their fees. These include upload fees, egress Fees, storage fees, retrieval fees, maintenance fees, and could also include other fees.

Upload Fees: The cost of uploading video footage from the customer's site to the cloud-based storage infrastructure can also add to the overall cost.

Egress fees themselves can be a significant cost associated with cloud-based storage. Egress fees are charged when video data is accessed or downloaded from the cloud storage provider. Some cloud storage providers do offer free or discounted egress fees for certain volumes of data or during certain times of the day. Customers should carefully evaluate egress fees to ensure they select the most cost-effective option for their video surveillance infrastructure.

For example, Amazon Web Services (AWS) charges \$0.09 per GB for data egress in their standard tier, while Google Cloud Platform (GCP) charges \$0.12 per GB for data egress.

Storage Fees: The cost of storing video footage

in the cloud-based storage infrastructure is another significant cost to consider. Most cloud storage providers charge for storage on a per GB or per TB basis, and the cost can vary depending on the provider's pricing policies and the volume of data being stored. For example, AWS charges \$0.023 per GB per month for standard storage in their S3 service, while GCP charges \$0.02 per GB per month for standard storage in their Cloud Storage service.

Retrieval Fees: The cost of retrieving video footage from the cloud-based storage infrastructure is another factor to consider. Retrieval costs are charged when the customer downloads or accesses data from the cloud. Retrieval costs can vary depending on the provider's pricing policies and the volume of data being retrieved. For example, AWS charges \$0.0004 per request for standard retrieval in their S3 service, while GCP charges \$0.01 per GB for retrieval from their nearline storage service.

Maintenance and Support Fees: The cost of maintaining and supporting the cloud-based storage infrastructure can also add to the overall cost. Maintenance and support costs can include fees for software updates, technical support, and other services provided by the cloud storage provider.

Other Pricing Issues

Retaining more and more video each day becomes a costly proposition in terms of both storage and egress fees. In traditional surveillance systems, video data is often recorded over in a matter of days or weeks.

However, with cloud-based storage, video data can be retained indefinitely, and the costs can quickly spiral out of control.

More Pricing Examples

According to Amazon Web Services (AWS), they charge \$0.023 per GB per month for the first 50TB of data stored. After that, the cost per GB drops to \$0.022. Google Cloud Storage charges \$0.020 per GB per month for the first 50TB of data stored, with the price dropping to \$0.010 per GB per month for data stored above 50 TB.

In addition to the cost of storing data, we need to remember those egress fees when data is transferred out of the cloud. Egress fees can quickly add up, especially when dealing with large video files. AWS charges \$0.090 per GB of data transferred out of the cloud, while Google Cloud Storage charges \$0.12 per GB. These fees can quickly add up, especially for businesses that may access large amounts of video data frequently.

Furthermore, cloud-based storage solutions can quickly become costly if businesses need to retain video data for extended periods. For example, if a business needs to retain video data for one year, the cost of storing 100TB of data on AWS would be \$27,600 (\$0.023 per GB per month x 100,000GB x 12 months). If that same business needed to access 10% of that data every month, they would be charged an additional \$108,000 in egress fees (\$0.090 per GB x 100,000GB x 10% x 12 months).

Another example, if a customer is using a 100Mbps network connection to transfer their video surveillance footage to the cloud. Assuming that

the footage is being uploaded continuously at a rate of 100Mbps, the customer would be transferring approximately 3.5Terabytes of data each day. If the customer is using a cloud-based storage provider that charges \$0.02 per gigabyte of data transferred, the daily cost of transferring 3.5Terabytes of data would be approximately \$70. Over the course of a month, this would result in a bandwidth cost of \$2,100.

It's also important to note that bandwidth costs can be unpredictable, as they can be affected by a variety of factors, such as network congestion or unexpected spikes in data usage. This unpredictability can make it difficult for customers to accurately predict their monthly costs and can result in unexpected charges.

Video Data Ownership and Security Issues

Another significant challenge of using cloud-based storage in video surveillance infrastructure is ownership issues. Customers should carefully evaluate the ownership of their video data once it is stored in the cloud. Customers should ensure that they retain ownership of their video data and that they have the right to access, download, and delete their data at any time. Furthermore, customers should ensure that their data is protected from unauthorized access, use, or disclosure.

Security is another major challenge of using cloud-based storage in a video surveillance infrastructure. The sensitive nature of video data makes it a prime target for cybercriminals,

and the risk of data breaches is significant. Moreover, cloud storage providers may not provide the same level of security as an on-premise storage solution, leaving the video data vulnerable to unauthorized access and tampering.

Additionally, security issue with cloud storage is the lack of control over the data. With on-premise storage solutions, the organization has complete control over the data and can implement their security measures to protect it. However, with cloud-based storage, the organization relinquishes control over the data to the cloud provider, who may have different security protocols and policies. Moreover, the organization may not have visibility into who has accessed the data or how it has been used, which can be a significant concern in high-security environments.

Customers must work with cloud providers that can offer the necessary support, flexibility, and cost-effectiveness required to make the transition to cloud-based storage successful.

Therefore, implementing a cloud-based storage infrastructure for a video surveillance system can involve various costs, including network connection costs, upload/egress fees, storage fees, retrieval fees, and maintenance and support fees. It's essential to consider all of these costs when evaluating options for a cloud-based storage infrastructure to ensure that the solution meets the customer's needs and budget.

And there is still the issue of easy video playback of all recorded video.

CHAPTER EIGHT

LTO Library Physical Connection



Installation Overview

As discussed throughout this book, in modern data-intensive environments, efficient data storage systems—especially data solutions such as video surveillance—are crucial. Linear Tape Open (LTO) technology offers the most efficient, reliable, and cost-effective storage resources available.

This chapter outlines the process of physically connecting an LTO library (some call these devices ‘jukeboxes’) to a PC server using a Serial Attached SCSI (SAS) Host Bus Adapter (HBA). Additionally, it covers the optional connection using a Fibre Channel (FC) HBA.

We will also explain how to set up multiple LTO library connections to a single server via the use of a hydra-cable, how to manage multiple LTO libraries, and how the connected library is recognized and utilized by the server.



An example of a LTO Library with multiple LTO Drives. The LTO Drive (or multiple drives) can support a large library with many LTO slots.

Components and Terminology:

- Linear Tape Open (LTO) Library: A device (sometime referred to as a “jukebox”) that houses multiple LTO tape drives and

slots that house LTO tape cartridges for data storage and retrieval.

- **PC Server:** A computer system that acts as a central processing unit for various tasks including managing the LTO Library and drives.
- **Host Bus Adapter (HBA):** A hardware component (typically PCI) that fits into the PC Server, which facilitates communication between the server and external storage devices.



Example of a SAS HBA

Connecting LTO Library to PC Server with SAS HBA:

Requirements:

Before you begin, ensure you have the following components and prerequisites:

- LTO library with SAS interface
- SAS HBA card
- Mini SAS cables
- PC server with available PCIe slots

- Compatible SAS drivers for the server's operating system

Instructions

- 1) Power off the PC server and the LTO library.
- 2) Open the PC server chassis and locate an available PCIe slot.
- 3) Insert the SAS HBA card into the server's PCIe slot and secure it with the appropriate screws.
- 4) Connect one end of the Mini SAS cable to the SAS HBA port on the server and the other end to the SAS port on the LTO library. Ensure a snug connection.
- 5) **SAS Cable Connection:** The cable has keyed connectors to ensure proper alignment.
- 6) Power on the LTO library first and make sure it is completely ready.
- 7) Power on the PC server.
- 8) Install the required SAS drivers for your server's operating system. These drivers can usually be downloaded from the HBA card manufacturer's website.
- 9) Once the drivers are installed, the SAS HBA card should detect the LTO library. You can verify this by checking the device manager or using the manufacturer's management software.
 - a) You may need to reboot your server for the drivers to properly load and take effect.
- 10) Initialize and configure the LTO library according to the manufacturer's instructions.



Example of a FC HBA

Connecting LTO Library to PC Server Using an FC HBA :

Requirements:

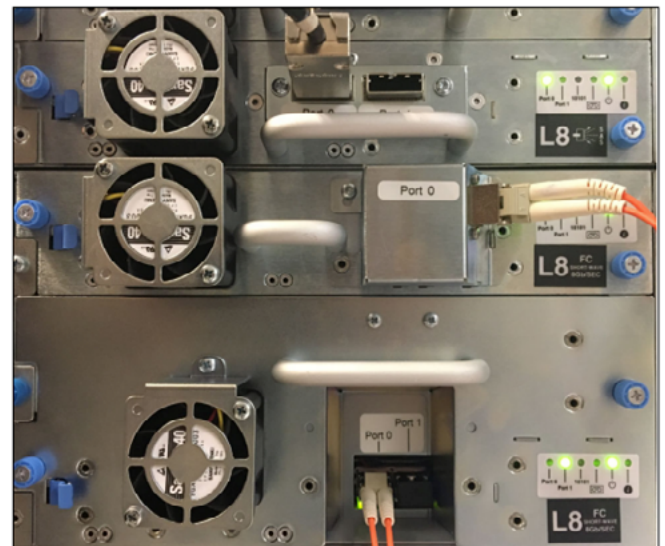
If you wish to add an FC HBA for enhanced performance and redundancy, you'll need the following additional components:

- FC HBA card
- FC cables
- FC switch (if connecting multiple devices)

Instructions

1. Power off the PC server and the LTO library.
2. Open the PC server chassis and locate an available PCIe slot.
3. Insert the FC HBA card into the PCIe slot and secure it with screws.
4. Connect one end of the FC cable to the FC HBA port on the server and the other end to the FC switch. If you're connecting directly to the LTO library, connect the cable directly.

5. Power on the LTO library, the PC server, and the FC switch (if applicable).
6. Install the necessary FC drivers for your server's operating system. These drivers can usually be obtained from the FC HBA card manufacturer's website.
7. Once the drivers are installed, the FC HBA card should detect the FC network and the LTO library.
8. Configure the FC HBA settings, including zoning (if using an FC switch), according to the manufacturer's guidelines.



Example of the back panel of an LTO library with FC connectors

Multiple LTO Library Connections to One Server

Connecting multiple LTO libraries to a single server allows for increased data storage capacity and redundancy. However, several factors need to be considered:



Example of a multi-connector 'hydra cable'

Hydra-Cable And Its Usage

A hydra-cable is a multi-connector cable that consolidates multiple SAS cables into a single cable.

It simplifies cable management and improves airflow within the server chassis. To use a hydra-cable:

1. Connect the individual SAS cables to the corresponding ports on the hydra-cable.
2. Connect the other end of the hydra-cable to the SAS HBA card.
3. Sufficient PCIe slots: Ensure your server has enough available PCIe slots for the additional HBAs.
4. Resource allocation: Plan for proper resource allocation, including CPU, memory, and bandwidth,

to accommodate multiple library connections.

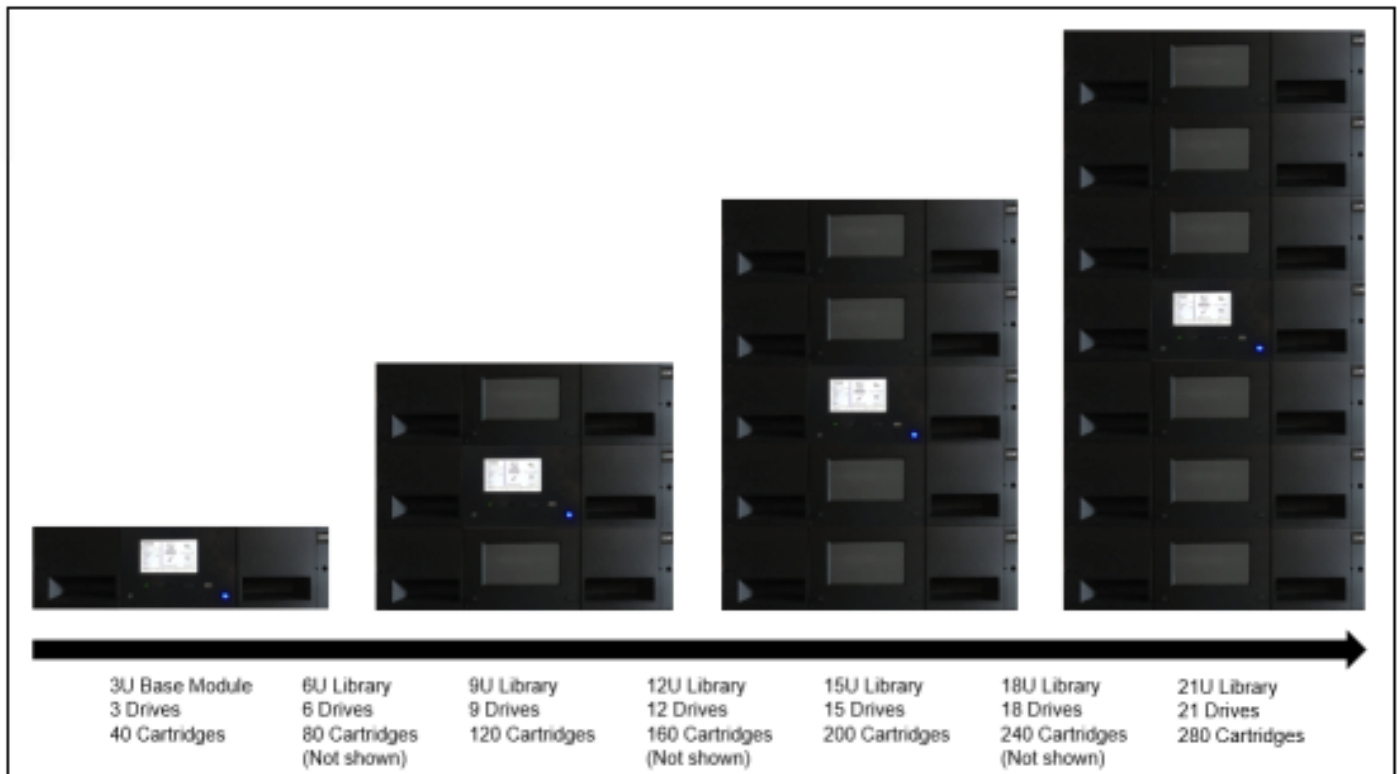
5. Operating system support: Check if your server's operating system supports multiple HBAs and libraries.
6. Cable management: Organize cables to prevent tangling and ensure proper airflow within the server chassis.

Instructions

1. Follow the instructions in sections 2 and 3 to install SAS and/or FC HBAs for each LTO library you want to connect.
2. Make sure each HBA is installed in a separate PCIe slot and connected to the corresponding LTO library.
3. Install the required drivers for each HBA and configure them according to the manufacturer's guidelines.
4. Initialize and configure each LTO library individually, ensuring that they are recognized and accessible by the server.
5. Depending on your server's operating system and backup software, configure backup jobs to utilize the multiple LTO libraries efficiently.

Connecting an LTO library to a PC server using a SAS or FC HBA is a critical step in establishing a reliable video storage solution.

By following the instructions provided in this chapter, you can ensure a secure and efficient connection, while also considering options for multiple library connections in order to best scale your video storage infrastructure to meet the demands of your video surveillance storage requirements.



Example of how to stack and scale LTO libraries for greater retention/storage

References and Resources For further information and guidance on LTO library connections, SAS HBAs, and FC HBAs, you can refer to the following resources:

- “LTO Technology Overview” by the LTO Program:
<https://www.lto.org/what-is-lto/>
- Linear Tape-Open (LTO) Consortium:
<https://www.lto.org/>
- SAS (Serial Attached SCSI) Protocol:
https://www.scsita.org/library/serial_attached_scsi_technology_roadmap/#
- Fibre Channel Industry Association (FCIA):
<https://fibrechannel.org/>
- Microsoft Docs - Device Manager:
<https://docs.microsoft.com/en-us/windows-hardware/drivers/install/device-manager>
- Microsoft Docs - Disk Management:
<https://learn.microsoft.com/en-us/windows-hardware/manufacture/desktop/add-device-drivers-to-windows-during-windows-setup?view=windows-11>
- Linux SCSI Generic (sg) HOWTO: <http://ltdp.org/HOWTO/SCSI-Generic-HOWTO/>



Example of the askALICE PC Server (top) integrated with the LTO Library (bottom) to provide a complete 2-Tier video surveillance storage solution.

CHAPTER NINE

— Calculating and Configuring a 2-Tier Video Surveillance Storage Solution



So far, this book has explained the capabilities and benefits of a 2-Tier video surveillance storage solution and in this chapter, we will walk through the steps to calculate those two tiers of video storage as well as best practices on configuring such a solution.

We will need to understand the various elements of a video surveillance solution to calculate the amount of video (data) storage needed. These include:

- a) Number of surveillance cameras:** This is the number of cameras in the system. More cameras mean more footage to store.
- b) Camera resolution:** This is the number of pixels in the video image. Higher resolution cameras produce more detailed footage but also require more storage space.
- c) Motion Activity:** This refers to the amount of

movement in the scene being monitored. More motion activity means more data to store.

d) Frame rate: This is the number of individual frames per second that the camera captures. Higher frame rates produce smoother footage but also require more storage space.

e) Number of Days Retention: What has-to-date- been more of a “assumed” or “settled for” retention period, with our 2-Tiered solution, you can calculate the amount of video data storage required based on how much retention your organization wants or would prefer.

The primary element in this overall calculation is the resolution of the cameras. Below is a chart with examples of the various surveillance camera resolution capabilities.

Term	Resolution	Aspect Ration	AKA Term	Resolution	Resolution Per Second	Resolution to Bits conversion
QCIF	176 x 120	4:03	QCIF	21,120	633,600	19,008,000
CIF	320 x 240	4:03	CIF	76,800	2,304,000	69,120,000
2CIF	704 x 240	4:03	2CIF	168,960	5,068,800	152,064,000
4CIF	704 x 480	4:03	4CIF	337,920	10,137,600	304,128,000
VGA	640 x 480	4:03	VGA	307,200	9,216,000	276,480,000
D1	720 x 480	4:03	D1	345,600	10,368,000	311,040,000
WVGA	752 x 480	4:03	WVGA	360,960	10,828,800	324,864,000
SVGA	800 x 600	4:03	SVGA	480,000	14,400,000	432,000,000
XGA	1024 x 768	4:03	XGA	786,432	23,592,960	707,788,800
720p HD	1280 x 720	16:09	720p HD	921,600	27,648,000	829,440,000
960p HD	1280 x 960	16:09	960p HD	1,228,800	36,864,000	1,105,920,000
WXGA	1280 x 800	16:10	WXGA	1,024,000	30,720,000	921,600,000
SXGA	1280 x 1024	5:04	1.3MP	1,310,720	39,321,600	1,179,648,000
HD+	1600 x 900	16:09	HD+	1,440,000	43,200,000	1,296,000,000
UXGA	1600 x 1200	4:03	1.9MP (AKA 2MP)	1,920,000	57,600,000	1,728,000,000
1080p HD	1920 x 1080	16:09	1080p HD	2,073,600	62,208,000	1,866,240,000
2K HD	2048 x 1080	16:09	2K Digital Cinema	2,211,840	66,355,200	1,990,656,000
QXGA	2048 x 1536	4:03	3.1MP (AKA 3MP)	3,145,728	94,371,840	2,831,155,200
WQHD	2560 x 1440	16:09	WQHD	3,686,400	110,592,000	3,317,760,000
QSXGA	2560 x 2045	5:04	5.2MP	5,235,200	157,056,000	4,711,680,000
2592*1944	2592 x 1944	4:03	5MP	5,038,848	151,165,440	4,534,963,200
3K HD	3072 x 1728	16:09	3K RAW	5,308,416	159,252,480	4,777,574,400
4K HD	4096 x 2160	16:09	4K Digital Cinema	8,847,360	265,420,800	7,962,624,000
20MP	5472x3648	16:09	20MP	19,961,856	598,855,680	17,965,670,400
8K HD	8192 x 4608	16:09	8K Digital Cinema	37,748,736	1,132,462,080	33,973,862,400

As the resolution goes higher the better the overall video quality becomes.

The resolution of the footage captured by these cameras has a significant impact on storage capacities.

The resolution of a surveillance camera refers to the number of pixels it can capture in an image. The higher the resolution, the more pixels there are, resulting in a clearer and more detailed image. High-resolution cameras are becoming increasingly popular in the surveillance industry, as they provide greater detail and accuracy in identifying individuals and objects.

To calculate the necessary storage capacity for a video surveillance system, you can use the following formula:

$$\text{Storage Capacity} = (\text{Number of Cameras}) \times (\text{Camera Resolution}) \times (\text{Motion Activity}) \times (\text{Frame Rate}) \times (\text{Length of Time})$$

For example, let's say you have a video surveillance system with eight cameras, each with a resolution of 1920x1080 (2.1 megapixels), monitoring an area with moderate motion activity. You want to capture footage at 30 frames per second (fps) and store it for 30 days. Using the formula above, the necessary storage capacity would be:

Storage Capacity = (8 cameras) x (2.1 megapixels)
x (moderate motion) x (30 frames per second) x
(30 days x 24 hours x 60 minutes x 60 seconds)

Storage Capacity = $8 \times 2.1 \times 1.5 \times 30 \times 2,592,000$

Storage Capacity = 11,356,800,000 megabytes or
approximately 11.4 terabytes

So, in this example, you would need at least 11.4 terabytes of storage capacity to store 30 days of footage from eight cameras with a resolution of 1920x1080, capturing footage at 30 frames per second in an area with moderate motion activity.

Let's consider another example. Suppose we have a surveillance system with ten cameras, and each camera captures footage at 1080p resolution. Assuming each camera records for 24 hours a day, we can calculate the amount of storage required per day as follows:

Assuming a bitrate of 4000kbps (kilobits per second), the storage required per camera per day would be approximately 57.47 gigabytes (GB). Therefore, the total storage required per day for all ten cameras would be 574.7GB.

In contrast, if we had the same surveillance system with ten cameras, but each camera captured footage at 720p resolution, assuming the same recording time and bitrate, the storage required per camera per day would be approximately 31.95GB. Therefore, the total storage required per day for all ten cameras would be 319.5GB.

As we can see from the example, the resolution of the camera has a significant impact on

storage requirements. In this case, the higher resolution cameras require almost twice as much storage space as the lower resolution cameras.

There are significantly larger resolutions available today such as 4K and 8K surveillance cameras.

Video Compression Techniques

H.264, also known as MPEG-4 Part 10 or AVC (Advanced Video Coding), is a widely used video compression standard that efficiently compresses video data without significantly sacrificing video quality.

The H.264 compression algorithm works by dividing each frame of video into multiple smaller blocks, which are then analyzed and compressed using various techniques such as motion estimation, transform coding, and entropy coding.

In motion estimation, the algorithm determines the motion of each block between frames and uses this information to reduce redundancy in the video data. Transform coding involves applying mathematical transforms to each block to further reduce redundancy, while entropy coding uses variable-length coding to compress the remaining data.

The result of H.264 compression is a highly compressed video stream that requires less storage space than uncompressed video data. This makes it an ideal compression technique for video surveillance, where large amounts of video data need to be stored for long periods.

H.264's impact on video surveillance storage is significant. By using H.264 compression, surveillance footage can be stored in smaller file sizes, reducing the amount of storage space required for video data. This means that more video data can be stored on the same amount of storage hardware, reducing overall storage costs.

Compared to other video compression techniques, H.264 offers several advantages. It provides higher compression ratios than earlier video compression standards such as MPEG-2 or MPEG-4 Part 2, which means smaller file sizes and less storage space required. H.264 also supports a wider range of video resolutions and frame rates than earlier standards, making it more versatile for use in various applications.

Other compression techniques that are commonly used in video surveillance include H.265/HEVC (High-Efficiency Video Coding), which provides even higher compression ratios than H.264 but requires more processing power, and MJPEG (Motion JPEG), which uses JPEG compression on each individual frame of video, resulting in larger file sizes than H.264 but potentially higher quality. Ultimately, the choice of compression technique depends on the specific needs of the surveillance system and the available hardware resources.

From our example above, if the original resolution calculated to a bitrate of 4000kbps, applying H.264 and having limited motion in the video, the bitrate could be reduced to 1105kbps. This is a meaningful reduction in the amount of storage capacity needed.

Frame Rates and Debates on What We Can See

The human eye is often described as one of the most complex and sophisticated organs in the human body. It allows us to see the world around us and take in the beauty of the natural environment. However, as technology has advanced, there has been an ongoing debate about what the human eye can see in comparison to video cameras that can record at 30 frames-per-second or higher.

The human eye has over 120 million light-sensitive cells that work together to form a clear image of the world around us. The retina, the thin layer of tissue at the back of the eye, is responsible for receiving light and processing it into electrical signals that the brain can interpret. This process is incredibly complex and involves the coordination of many different cells and structures in the eye.

In terms of frame rate, the human eye does not actually see in frames per second like a camera or video recorder does. Instead, the eye constantly sends a stream of information to the brain, which processes it into a seamless, continuous image. However, studies have shown that the human eye can distinguish between separate images presented at a rate of up to about 60 frames per second (fps) or higher under certain conditions.

It's important to note that the perception of motion is affected by various factors such as the complexity of the scene, the brightness of the image, and the individual's visual acuity. Therefore, the exact frame rate at which a person can perceive smooth motion may vary.

Studies on frame rates and the human eye can be studied with articles such as “How the Human Eye Works” by Tim Sharp, Live Science, April 16, 2018, or “The Frame Rate of Human Vision” by Dr. Peter J. Bex, Current Biology, Volume 26, Issue 22, R971-R972, November 21, 2016.

One argument in favor of the human eye being able to see more than video cameras is the concept of dynamic range. Dynamic range refers to the range of light levels that a camera or the human eye can perceive. While video cameras are limited in their dynamic range, the human eye is capable of adapting to a wide range of light levels, from bright sunlight to dimly lit rooms. This means that the human eye is better equipped to see detail and color in both bright and dark areas than video cameras.

Another argument in favor of the human eye being able to see more than video cameras is the concept of resolution. While modern video cameras are capable of capturing high-resolution images and videos, the human eye is still able to perceive much more detail than even the highest-resolution cameras. This is because the human eye is capable of focusing on different areas of an image and adjusting its focus to create a sharper and more detailed image.

However, there are also arguments in favor of video cameras being able to see more than the human eye. One of the main arguments is that video cameras are capable of capturing a much wider field of view than the human eye. While the human eye has a field of view of around 180-degrees, video cameras can capture a field of view of up to 360 degrees. This means that video cameras are capable of seeing things that the human eye cannot, such as events happening behind the viewer.

Why We Use Slow Motion Video

Slow-motion recording is the process of capturing motion at a slower speed than it actually occurred. This effect is achieved by recording the video at a higher frame rate than the normal playback rate and then playing it back at a standard frame rate. For example, if you record a video at 240 frames per second (fps) and play it back at 24fps, the resulting video will be slowed down by a factor of 10. This means that one second of real-time motion will be stretched out to 10 seconds of slow-motion playback.

The higher the frame rate used in recording, the more detail will be captured. This is because at higher frame rates, more individual frames are captured per unit of time, which means that the motion is recorded with greater temporal resolution. As a result, the slow-motion playback will reveal more details that are not visible at normal playback speeds.

For instance, recording a hummingbird in slow-motion at a high frame rate of 1000 fps would capture the wing movements in detail, revealing the rapid fluttering that is not visible to the naked eye. Similarly, recording a golf swing at 480 fps would capture the precise movement of the club and the ball, allowing for a more detailed analysis of the swing mechanics.

Slow-motion video recording is a commonly used technique in video surveillance in the casino gaming industry, especially in games like blackjack. It allows the surveillance team to capture and review critical moments of the game, including player and dealer interactions, card shuffling, and any suspicious behavior or cheating.

For instance, in blackjack, slow-motion video recording can help the surveillance team detect various types of cheating, such as card marking, dealer collusion, or chip manipulation. It can also help identify players who are counting cards, which is not technically illegal but can lead to ejection from the casino if caught.

One example of how slow-motion video recording can be useful in blackjack surveillance is by capturing the moment when a player tries to switch cards. Suppose a player swaps a low-value card with a high-value card from their sleeve during the game. In that case, slow-motion recording can help the surveillance team detect this type of cheating and take appropriate action, such as ejecting the player from the casino.

Another example is using slow-motion video to analyze how a dealer shuffles the cards. If a dealer is not shuffling the cards properly or is using sleight of hand to manipulate the deck, slow-motion recording can help identify this behavior and prevent cheating.

Overall, slow-motion video recording is an essential tool for casino surveillance teams to ensure fair play and maintain the integrity of the games.

Therefore, the argument for recording video at higher than 30fps is becoming more important than recording at lower than 30fps because of video storage issues.

Calculating A Real-World Video Surveillance Solution

Let's walk through a potential real-world example where a 2-Tier video surveillance storage solution would also have a meaningful difference, especially in the acquisition costs of the equipment.

a) Number of Surveillance Cameras: 50

b) Camera Resolution: 1080p

c) Motion Activity: Average amount of motion in the video over time: 30%

d) Frame Rate: 30fps

e) Number of Days Retention: 180-days

With H.264 compression and 30% motion, these variables would generate a bitrate of 3110kpbs. And using the formulas from above we see that this example would require 281.53-Terabytes of needed video data storage.

In a traditional single tier storage solution, a 32-bay server using 10TB hard drives in a RAID-5 setup would be needed.

In our 2-Tier infrastructure and working with our video operator, we will create a first tier of hard-disk based storage at 10% of the overall needed storage, or 28Terabytes (TBs). Providing for 'overhead' and proper RAID data protection, let's make it a 30TB first tier.

This could be configured in a 4-bay network attached storage (NAS) servers using 10TB hard drives in a RAID-5 setup. And then a single 2U LTO library with capacity for 24 slots and only needing to use 16 LTO-9 cartridges.

Looking at the equipment acquisition costs as well as energy costs, we are looking at a 45% overall costs savings of the 2-Tier solution over that of a traditional storage solution.

You can input your own variables to determine what your costs savings could be at the FUJIFILM storage calculator:

<https://www.fujifilm.com/us/en/business/data-storage/resources/video-surveillance-retention-cost-calculator>

This chapter should reinforce the overall costs savings when utilizing LTO technology in a 2-Tier video storage solution and this e-book should highlight how a 2-Tier aware video management software (VMS) system can deliver the ease of to record, store, and manage surveillance video and effortlessly playback any and all recorded video.

CHAPTER TEN

Summary of the 2-Tier Video Surveillance Storage Concept



Recapping what we have covered in this book, as video surveillance systems continue to become increasingly sophisticated and data-heavy, the need for efficient and cost-effective storage solutions becomes more critical. In response to this growing demand, the 2-tier storage architecture utilizing Linear Tape-Open (LTO) technology has emerged as a highly viable option for video surveillance storage.

If you are looking to save significant amounts of money on video surveillance storage while still maintaining reliable, high-quality recordings, 2-tier storage with LTO technology is definitely worth considering.

Here are some of the key benefits and advantages of 2-tier storage with LTO technology:

Cost-Effective: One of the most significant advantages of 2-tier storage with LTO technology is the cost savings it provides. By using high-capacity LTO tapes for long-term storage and high-speed disk storage for immediate access, you can reduce the overall cost of your storage infrastructure. Additionally, because LTO technology is a highly stable and reliable storage medium, you can significantly reduce the costs associated with data loss or corruption.

Scalable: As your video surveillance storage needs grow, you need a storage solution that can keep pace. The 2-tier architecture with LTO technology allows you to easily scale up your storage capacity as needed by adding more tapes or disk storage devices. This scalability also makes it easy to upgrade your system without having to replace the entire infrastructure.

High-Speed Access: While LTO technology is excellent for long-term storage, it is not as fast as disk storage when it comes to accessing data. By utilizing a 2-tier storage architecture with LTO technology, you can take advantage of the high-speed access provided by disk storage for immediate access to recordings while still maintaining the reliability and longevity of LTO tapes for long-term storage.

High Capacity: LTO technology is well-known for its high capacity, with the latest LTO-9 tapes offering up to 18TB of uncompressed storage. By using LTO tapes for long-term retention, you can store massive amounts of video surveillance data in a compact, cost-effective form factor.

Energy Efficient: Unlike HDDs that are constantly spinning 24/7 consuming a lot of energy and generating heat that needs to be cooled, tape cartridges only consume energy when being read or written by a tape drive. Otherwise, most of the time a tape cartridge is not consuming any power as it rests in a tape library slot or on a shelf in an offsite vault. For this reason, studies show that over time, tape systems consume 87% less energy than equivalent capacities of HDD. Over the product lifecycle including raw materials, production, distribution, usage and disposal, tape system produces an astonishing 97% less CO₂e than HDD. This helps organizations meet increasingly vital ESG goals in light of climate change and global warming.

Data Security: LTO technology is designed to provide high levels of data security and reliability. With features like error correction and write-once-read-many (WORM) capabilities, you can be sure that your video surveillance data is safe and secure.

Longevity: Unlike other storage media, LTO tapes have a lifespan of up to 50 years. This makes them an ideal choice for video surveillance storage, where long-term retention is often required.

Overall, the benefits of 2-tier storage with LTO technology make it an excellent choice for video surveillance storage. It provides a cost-effective, scalable, and reliable storage solution that can meet the needs of even the most demanding video surveillance applications.

Looking to the future, we can expect to see continued advancements in both 2-tier storage architecture and LTO technology. For example, advancements in LTO technology are already making it possible to store even more data on a single tape, with the potential to reach capacities of 576TB in the future.

Additionally, advancements in video management software (VMS) applications are making it easier to manage and utilize 2-tier storage architectures. With LTO-aware VMS systems, administrators can seamlessly manage data across multiple tiers of storage, making it easier to optimize performance and cost.

Finally, we can expect to see increased adoption of cloud-based storage solutions in the video surveillance industry. While LTO technology will continue to be an essential part of many video surveillance storage solutions, cloud-based storage offers unique advantages such as remote access, offsite backup, and flexible pricing models. However, cloud based solutions often result in substantially higher TCO and are often targets for cyber criminals.

Try 2-Tier Storage with LTO Technology for Significant Savings

Organizations that are struggling with the cost and complexity of storing video footage should consider 2-tier storage utilizing LTO technology. This approach can help organizations to realize significant savings while still ensuring that video footage is available when needed.

To encourage organizations to try 2-tier storage with LTO technology, the following steps can be taken:

1. Start small: Organizations can start by implementing a pilot project to test the feasibility and effectiveness of 2-tier storage with LTO technology. By starting small, organizations can minimize risk while still gaining valuable insights into the benefits and advantages of this approach.
2. Evaluate existing storage infrastructure: Before implementing 2-tier storage with LTO technology, organizations should evaluate their existing storage infrastructure to identify areas where improvements and cost savings can be made. This evaluation should include an analysis of storage capacity, performance, ease of use (video playback), and cost.
3. Partner with a trusted vendor: Organizations should partner with a vendor that has experience in implementing 2-tier storage with LTO technology. This partnership can help organizations to overcome any technical challenges and ensure that the storage solution meets their specific requirements.
4. Train staff: Organizations should provide

training to staff on how to use the new storage solution effectively. This training should include best practices for data management and VMS operator playback capabilities. Utilizing a system that allows for the VMS operator to simply move the VMS timeline to playback any and all recorded video is paramount in a proper 2-tiered video surveillance solution.

WHITE PAPER:

California Municipality Needs To Keep Video Surveillance Retention For One Year: On A Budget



The Challenge

California's state records management requirements dictate retaining certain government-created documents for a period of one year. And upon verification, video surveillance files are considered government records and certain entities are obligated to retain recorded video for a period of at least one year.

Retaining video surveillance recordings on multiple video surveillance cameras within a government agency could be a daunting task, both financially and technologically, over an entire year.

For the City of Oceanside and specifically the San Luis Rey Wastewater Treatment Plant – which has over two dozen video surveillance cameras around the facility– the importance of long-term video retention was not only

regulatory but also necessary for the security of the facility.

Keeping a city's water treatment facility safe and secure is a serious matter to numerous stakeholders. Therefore, these surveillance cameras are 1080p high-def IP-cameras and are recording 24/7 with the City connecting another 14 surveillance cameras from another location for more than 30 camera feeds.

With the need for such long-term video retention on that many surveillance cameras, the City of Oceanside faced the challenge of having the ability to easily review all recorded video and being able to afford such retention.

Needing to work within the set City budget allotments, the City's I.T. team needed a solution that would deliver the best video recording and at the same time meet the appropriate state's record retentions requirements.

Reviewing the video storage requirement for those over 30 surveillance cameras, one year's worth of retention would need approximately 548-terabytes of video storage. A traditional hard-drive only storage solution would be simply too expensive for the City's budget.

The physical security industry has tried numerous ways to find ways to store large amounts of video without taking up so much disk space. Concepts such as H.264, H.265, and H.265+ are the most popular. The introduction of H.264 and H.265/+ has significantly helped in reducing the amount of needed storage for video feeds. This video algorithm has the ability to maintain the level of quality video while saving a substantial percentage of storage space.

The challenge with H.264/H.265 is that as the number of actual cameras increases within any organization, the need for more and more storage space still arises and we are back to our vicious storage cycle.

Vendors also introduced the concept of "Record-on-Motion" through which the software would only store video "where there is activity." But then we had issues with trees moving in the background or birds flying through the scene, or lights flickering that cause "motion" but didn't have any "real" activity. So, the vendors allowed for "thresholds" of activity to be ignored. All this did was cause systems to NOT record and store video when something important DID happen. But at least it saved storage space.

Although overall helpful, these work-arounds have been focused on ways to save hard disk storage space to reduce the overall cost of the solution.

The City of Oceanside Water Treatment Plant needed a better, more cost-effective, affordable, solution that would provide them with the video (records) retention requirements and those guidelines as set forth by the City.

The Solution

askALICE from Cozaint.



The City learned about Cozaint's professional video surveillance management system with 2-tier video storage.

"We needed to upgrade our Security Surveillance system at our water treatment plant and since we have worked with several different systems, we knew what we liked and what we didn't like. It is always difficult to find a system that fits within our budget and meets our long-term retention requirements. Cozaint's askALICE has been an ideal fit for our surveillance recording needs and our budget." Scott Prather, Senior IT Analyst, City of Oceanside, CA.

Linear Tape-Open (LTO) Ultrium is a high-capacity, single-reel data tape storage solution developed and continually enhanced by Hewlett Packard Enterprise, IBM, and Quantum and promoted by the LTO Program.

First introduced in 2000, LTO technology is currently in its 9th generation. LTO-9 specifications support a cartridge with a capacity

of 18-terabytes and data transfer rates of up to 400MB per second for over 1-Terabyte of storage performance an hour per drive.

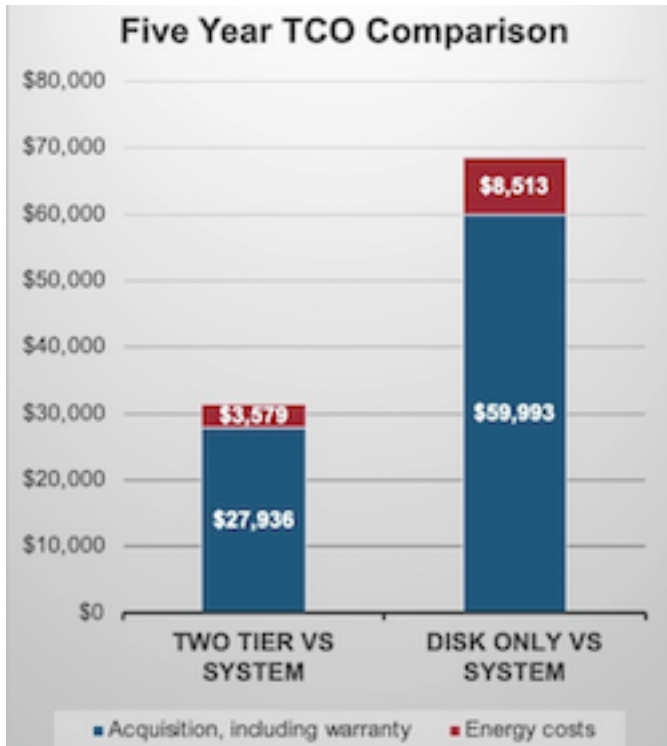
As hard disk-based storage solutions become very expensive in large capacities, what other storage technologies exist that could scale to such capacities?

Benefits of LTO:

LTO is a powerful, scalable, and adaptable data tape format that helps address the growing demands of long-term video retention. LTO Data Tape remains unrivalled in terms of cost for capacity, reliability, portability and security and it continues to play a crucial role in data protection. And LTO perfectly complements a more responsive disk-based storage system by providing a nearline storage offering.

Add LTFS. The Linear Tape File System (LTFS) makes viewing and accessing tape files easier than ever. LTFS works in conjunction with LTO technology for ease of use and portability for open systems data tape storage. With LTFS, one partition holds the content and the other holds the content's index, so the tape cartridge can be self-describing to improve archive management.

The costs were calculated based on data entered into the FUJIFILM Total Cost of Ownership (TCO) website tool (<https://www.fujifilm.com/us/en/business/data-storage/resources/video-surveillance-retention-cost-calculator>) made from the City of Oceanside water treatment plant usage and needs.



Storing More Video For Less Costs:

Savings for using LTO tape as the 2nd-tier of storage and augmenting a smaller disk capacity for video surveillance storage came to 54%. Less than HALF the cost of a traditional hard-disk only solution.

Putting It All Together – askALICE

An ideal video storage infrastructure would provide medium- and long-term storage and:

- **It Should Provide** easy access to all stored video;
- **Video Quality** should never be degrade, either initially or over time;
- **There should be NO extra steps** necessary to playback any/all recorded video
- **It should easily scale** as needed to support more surveillance cameras.

The ideal solution is a combination of disk-based storage and LTO-based storage in a two-tier solution. In such a multi-tiered storage infrastructure, a much smaller amount of hard disk storage is used as an initial, fast response, storage tier with the more cost effective LTO storage tier utilized for complete and long-term retention.

Cozaint has taken the requirements of such an affordable network video recording system and put together the appropriate software and hardware pieces to fulfil the needs of most medium to large scale video surveillance installations.

Combining specialized off-the-shelf PC hardware running the innovative BOBBYvms surveillance software and LTO libraries, Cozaint has come up with one of the most economical, flexible, scalable long-term video retention solutions on the market.

The BOBBYvms software comes with modules to add cameras, set up and modify user defined rules to direct the flow of video data, observe and control any camera in the system, maintain the video storage, and track the flow of video to the LTO library. In addition, the video for specific cameras is kept on the same LTO tape meaning that recall of the video for a particular camera is very efficient and effortless.

Cozaint's askALICE solution affordably scales from hundreds of terabytes to petabytes of video retention. Let us show you how we can deliver the long-term video storage you need at costs significantly lower than any other disk-only solution.

Total Cost Of Ownership (TCO) Results

	Two Tier VS System	Disk Only VS System
Tape Drives, Library, Media	\$20,618	\$0
Disk Cache	\$7,318	\$59,993
Energy	\$3,579	\$8,513
Cloud service fees + bandwidth	\$0	\$0
Solution 5 Year TCO	\$31,515	\$68,506

The Results

Cozaint's askALICE solution has been able to deliver the video surveillance long-term retention needs within the budget of a City government. askALICE is able to easily meet records retention policy needed by the City, while supplying the City's IT and Security team with the ability to quickly and easily review all recorded video.

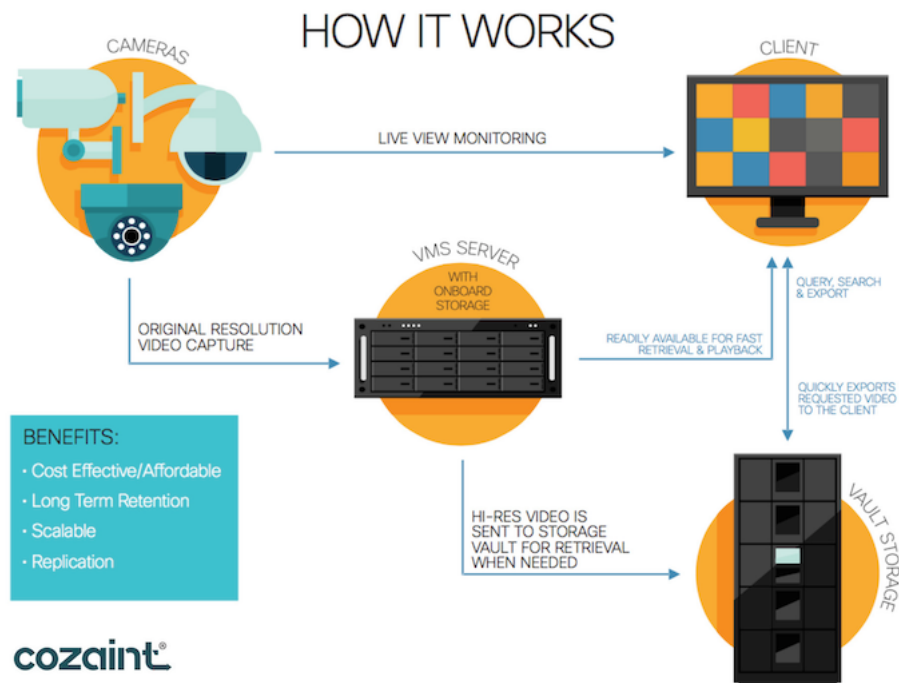
The ability for the video surveillance operator to easily playback any recorded video from any time period, without any extra steps or individuals, is what delivers the overall value of this two-tiered video storage solution.

The reliability of the askALICE solution has proven itself with being operational for over one-year already at the San Luis Rey Wastewater Treatment Plant. Cozaint was able to deliver a technical solution—where the IT team is unfamiliar with the daily operations of the security team and their need to quickly access and review video—can be challenging at best. With askALICE, the security team can interact with all recorded and stored video with no interaction from the IT team. And this saves operational costs.

The City of Oceanside understood how the traditionally high-cost of video storage retention would typically reduce the length of time -or even the

quality of the recorded video- the City could afford to save. And to keep the quality of video for the longest period of time desired, the City needed the flexible, cost effective, askALICE solution.

"The askALICE solution has provided the features and functionality our team needs from a video surveillance solution together with the long-term retention needs required by our City policies," added Prather.



This is an overview of how the askALICE 2-Tier Video Surveillance Storage solution works from camera feed ingest, to first tier recording, to second tier replication, to VMS operator playback of any recorded video.

Hollywood Comes To Video Surveillance

Dailies. B-Rolls. Circle-Takes. These digital video oriented processes of the “Hollywood” production marketplace have –for over a decade now– seen an explosion in the volume of recorded video that must be stored and managed. Entirely new workflows have been created to handle the deluge of video that digital movie-set cameras have unleashed. In the old days parts of movies, TV shows, and commercials would end up on the “cutting room floor” as sections of film were edited out of the production. Nowadays, every “take” is kept and possibly re-used in the bloopers edition or the director’s cut release.

SPOILERALERT: This article is about the technical management and storage of surveillance video and not an article on the cool special-effects Hollywood is pretending to do with video surveillance.

What our colleagues in the very similar Media and Entertainment (M&E) –or Hollywood– marketplace have learned is how to manage this vast amount of (and significantly growing) recorded video that is generated every day. How do directors and producers quickly access and review today’s shoot? How do they select the takes and scenes that make it into a movie? How can they quickly and easily find video scenes previously recorded for other productions and reuse them in a new production? What’s the most cost effective, affordable, way to store all of



these video assets? These are all questions the Hollywood marketplace has already figured out how to answer.

Therefore, with the IP video surveillance marketplace managing such a similar process to Hollywood in terms of storing vast amounts of recorded video, how come we have not embraced similar workflows in surveillance video lifecycle storage? Hollywood has already learned how to monetize recorded video over and over again. Hollywood has already learned how to use tried-and-true I.T. storage technologies to store terabytes and petabytes of video at the lowest possible costs (especially operational costs).



Hollywood has already learned how to add more information (metadata) to the recorded video to help make it more relevant for quicker searches in the future. Hollywood has already learned that if you can't get back to the recorded video quickly and easily, it quickly becomes useless to retain it.

Many people in the M&E market will point to the introduction of the RED digital camera as the tipping point for Hollywood to move to a digital workflow. Originally conceived in 2005, RED became a driving force in moving away from film and now so much is recorded on digital medium. Removing the high cost of movie and television grade film allowed for production companies to "save it all" and not leave any video footage on the proverbial cutting room floor. One small side effect: there is A LOT of extra video.

Let's examine two areas that Hollywood has gotten it right with regard to video management

and video storage. The first is metadata. Metadata (data about the data) allows each user to add information about the video that can be later used to easily search and review the video.

"Airplane flyover of Golden Gate Bridge at sunset" is a great example of a clip of video that a production company could easily reuse for that movie sequel about San Francisco. Airplane, golden gate bridge, and sunset are all terms (metadata) that would significantly help an editor quickly find this scene and use it again. Think of the cost savings this has over sending out another film crew to shoot another airplane flying over the famed bridge.

Similarly, being able to pinpoint the specific clip of video in a video surveillance environment becomes easy using



what existing digital storage technologies are trustworthy enough to store our video assets and ensure they will be there when we need them? Enter LTO digital computer data tape.

LTO storage has been available since 2000 and has become the de facto standard in computer data tape storage and is heavily used in the Hollywood marketplace. The challenge for many in the video surveillance / security marketplace is that when they see the word “tape” they hear “VHS.” Although tape is indeed a four-letter word, if you continue reading this article with such a feeling you will miss out on all of its benefits.

IBM, HP and Quantum developed LTO to counter other data tape technologies thus introducing a more open format. Much of the technology is an extension of the work done by IBM at its Tucson lab during the previous 20 years with over 80% of the world’s data residing on data tape.

Around the time of the release of LTO-1, Seagate’s magnetic tape division was spun off and eventually acquired by Quantum. Today, IBM, HPE, Quantum, Spectra Logic, Oracle, along with a number of others manufacture LTO data tape libraries with IBM manufacturing the LTO drives. The current shipping generation, LTO-9, at 18-Terabyte cartridges with a current street price of about \$125 per cartridge.

Adoption of LTO is already firmly established in the Hollywood media production environment. One of the driving forces behind

metadata. Metadata literally becomes worth its weight in gold.

Notice the similarities of the Hollywood market and the video surveillance market, especially between the Hollywood markets use of ‘metadata’ and surveillance industry use of video analytics.

The other area that Hollywood has learned how to do it right is in the use of multiple tiers of storage for the recorded video.

Just like in the video surveillance marketplace, video is not frequently reused or viewed after initial recording. And just like the video surveillance market, this rapid expansion of video assets is a relatively recent change and these newer solutions are now maturing. Therefore,

this adoption is a mandate to many feature motion picture productions by insurance companies that content (video) captured on set or on location be archived to LTO tape on a daily basis. LTO meets the dual needs of the studios and the insurance bonding companies. The bonding companies feel safer and more at ease with digital because the content is archived on LTO, the same tape-based platform that banks use.

LTO is rated at up to 50 years archival shelf life. It provides for 5,000 cartridge loads/unloads and it allows for approximately 260 full file passes (with one full pass equal to writing enough data to fill an entire tape cartridge). And with the sequential data structure format of video, LTO data tape becomes an ideal storage medium.

An appropriate surveillance video workflow however must be put into place to properly utilize LTO as a video surveillance storage medium. Just like in the Hollywood market, there needs to be a way to reference the original video stored on LTO data tape without any prolonged delays. Just like Hollywood's video editors, video surveillance operators cannot afford to wait to search and playback recorded video.

It is extremely important that the video surveillance management software provide the ability for the operator to move about the recorded video "timeline" without concern for where the video is ultimately stored.

Surveillance operators will need the ability to view any and all recorded video to find the proverbial needle in the ever growing video storage haystack.

As in almost every video surveillance investigation, there is a fair level of forward or backward "scrubbing" to actually find the video of interest we are looking to review. This is another reason the surveillance operator must have the ability, without any extra steps or intervention from the "I.T." staff.

Then, just like a Hollywood editor, a surveillance video operator can select the snippet of video needed to deliver to HR/Police/Court/etc by directing the system to the specific original, untouched – unaltered, recorded video located on a specific LTO cartridge. This taking but a few minutes of transfer time to complete. This best-practices approach to multi-tiered video storage also delivers appropriate "Chain-of-Custody" needed to submit the video as evidence in a court of law. However, it's the speed that surveillance video operators will care about most with the ability to quickly find the video they are interested in.

From a cost standpoint, this becomes a more superior infrastructure. Compare the acquisition costs of terabytes and petabytes of spinning NAS and SAN and then the cost of an appropriate LTO data tape library, complete with robotics. The LTO implementation is significantly less expensive.

Adding in the 3- to 4-year lifespan of spinning disks compared to the 50-year longevity of LTO incurs another maintenance cost. Now, throw in the ongoing monthly operational costs of electricity for all that spinning disk and the cooling costs to keep those SAN/NAS units spinning. 24/7. Year round. Those costs add up quickly.

Implementing a sound multi-tiered video storage infrastructure with spinning disk and LTO stored –unaltered– original video allows for more video to be stored for less. How do we create a disk-and-LTO infrastructure to provide us what we need in video storage, without breaking the bank?

When contemplating a 21st century surveillance video management solution, we realize that a spinning disk-only approach harkens us back to pre-Columbus days. There are better ways to reach our destination. We all want to store as much video as we can. We want to record at the highest resolutions. We want to record with the highest frame rates. And we want it as affordable as possible. So why not learn from our colleague in the Hollywood market?

I am confident that the video surveillance market will indeed adopt these workflow concepts from our colleagues in Hollywood and you will see increasing usage of LTO storage devices. Embracing the sequential data structure format of video along with the utilization of LTO data tape delivers an ideal storage solution. And done properly the implementation of LTO multi-tiered storage surveillance video recording is actually the perfect “killer app.”

Thanks Hollywood.

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Product Showcase: askALICE™

Cozaint askALICE

— Affordable Scalable Video Surveillance Storage



VMS Integration

Professional-grade enterprise-class powerful video management software



Scalable Disk and Cartridge Storage

Grow your video storage infrastructure as your need



Easy To Use

Simply move the VMS timeline to playback any and all recorded video



Most Affordable

Get ALL the video storage you want because you can afford this solution

Contact the Cozaint team at to go over your wants and needs for an affordable video surveillance storage system and learn how the askALICE system can save you on acquisition and total cost of ownership costs.

**<https://cozaint.com/contact-2/>
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askALICE™

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